Evolution of Cosmic Structure

Lecture 17: Galaxy groups and clusters: global properties

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Course Content

Introduction – Overview [1 lecture]
Background + Universe before decoupling [2 lectures]
Linear growth of perturbations [2 lectures]
Galaxy clustering and peculiar velocities [2 lectures]
Non-linear evolution of structure [2 lectures]
Numerical studies of structure formation [1 lecture]
Observations of the distant Universe [2 lectures]
Reionisation [1 lecture]
Galaxy formation and galaxy properties [3 lectures]
Host galaxies of supernovae [1 lecture]
Groups and clusters of galaxies [2 lectures]
Fate of the Universe/Galaxies [1 lecture]

Background and linear growth

Observational probes of linear growth and non-linear evolution

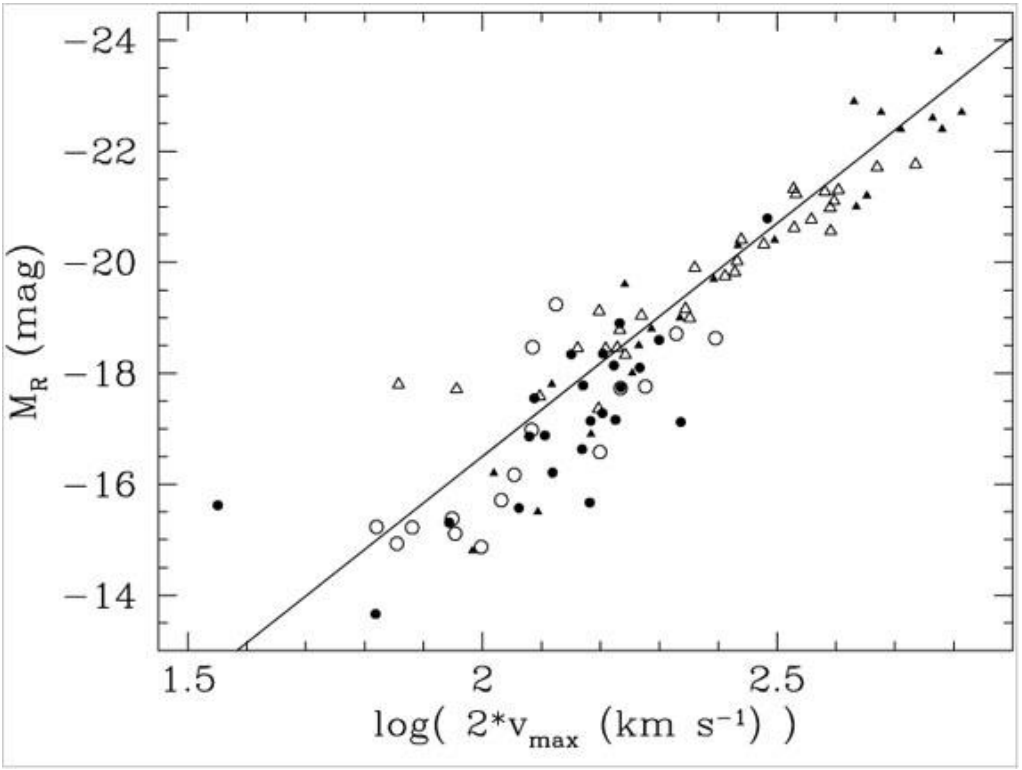
Galaxy Formation and Evolution across cosmic time

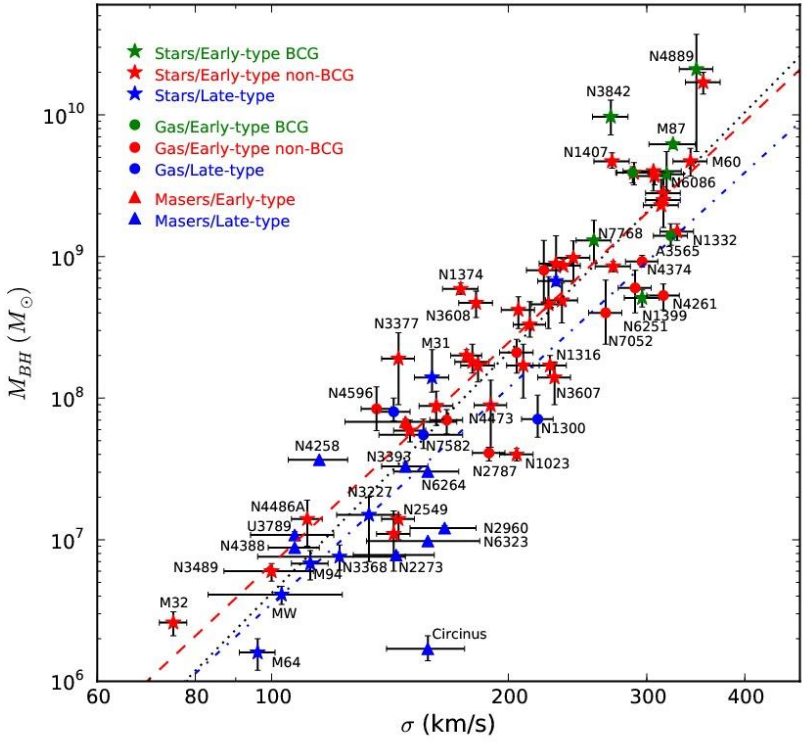
Galaxy groups, clusters and our fate

Starting Part 4 of the module

Recap: Galaxy Scaling

- Scaling relations for spirals (Tully-Fisher) and ellipticals (Fundamental Plane)
- Galaxy-BH co-evolution leads to relation between mass and velocity dispersion





Learning Outcomes

What are the contents and key properties of the largest collapsed structures in the Universe at the present time?

How do galaxies in clusters differ from the field, and what environmental processes are at work?

What do we learn from the properties of the hot intergalactic gas?

What scaling properties do groups and clusters follow, and how can they be understood?

How can the dominant dark matter be traced? What happens when clusters merge?

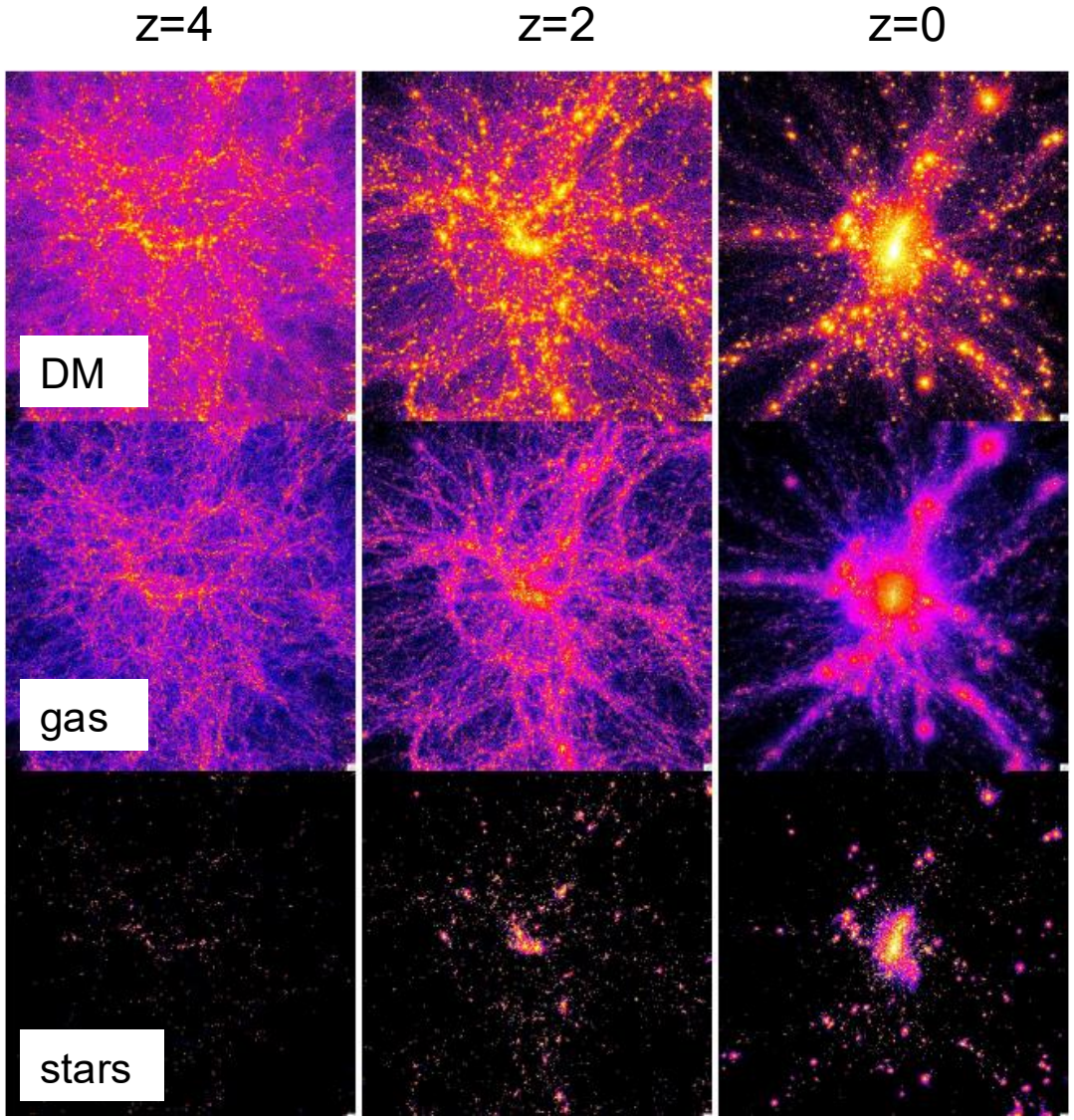
- Global properties and how do we find groups/clusters?
- Intergalactic gas and scaling properties
- Dark matter
- The evolution of clusters
- Galaxies in clusters

Galaxy groups and clusters

Cosmological simulations show clusters developing at the nodes of networks of cosmic filaments.

Dark matter halos form hierarchically, and gas concentrates into these.

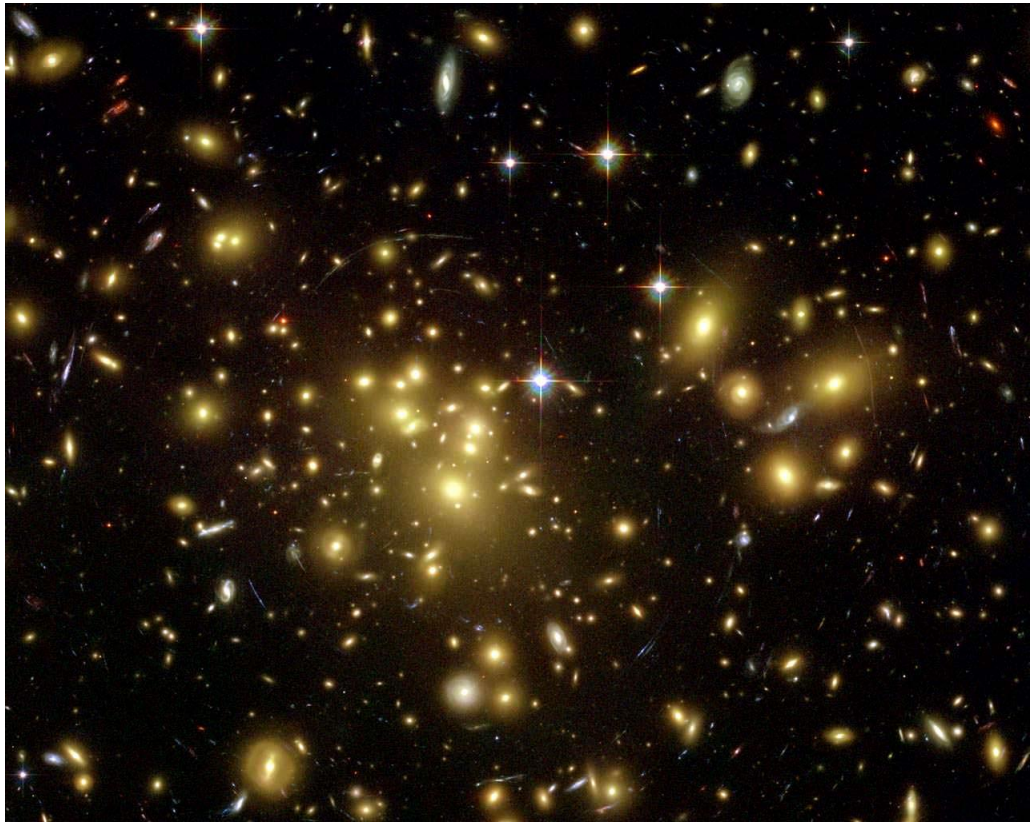
In small halos it is able to cool and form stars. In cluster-sized halos the virial temperature is ~10^7-10^8K, and cooling is less efficient.

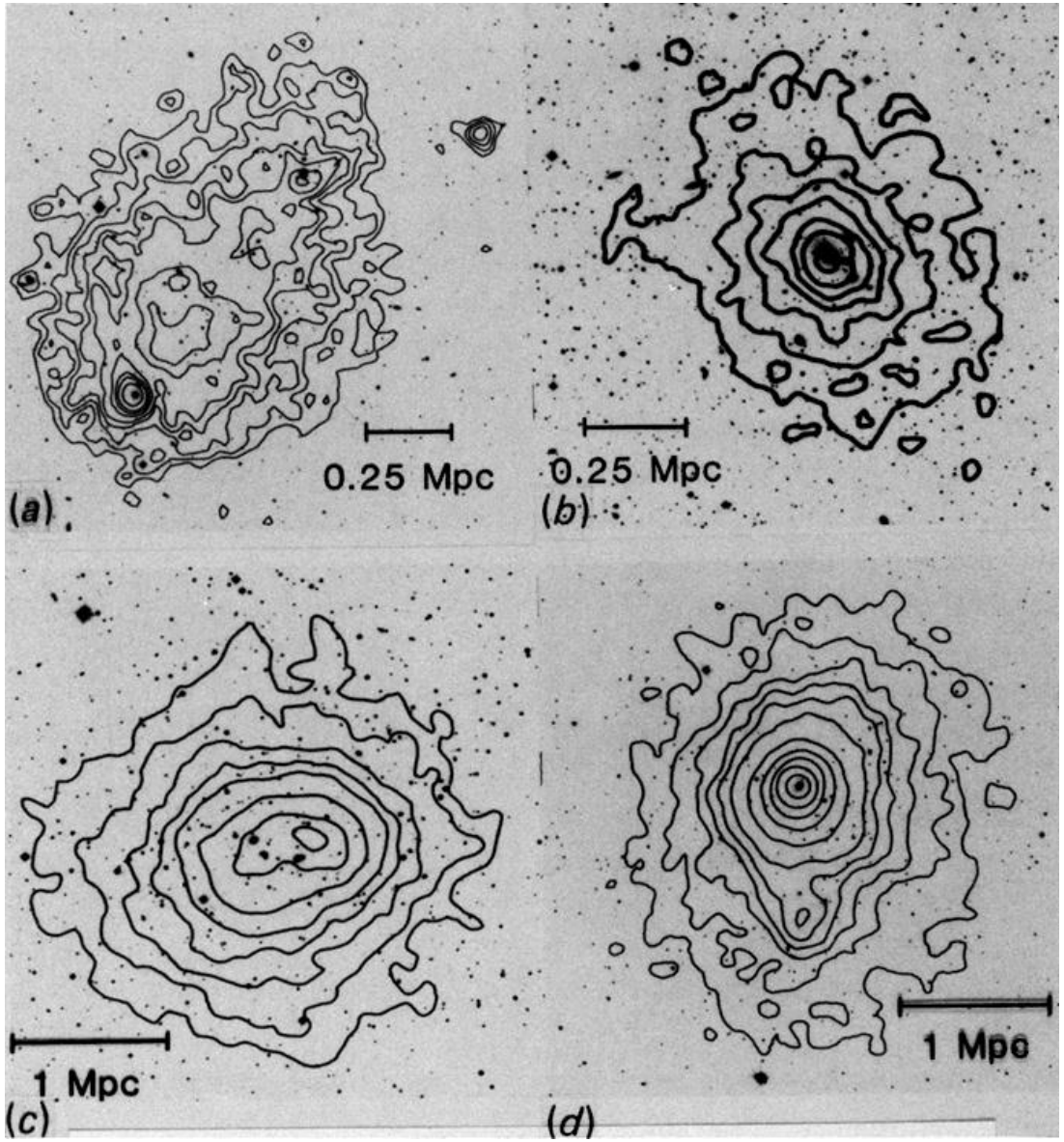


Borgani & Kravtsov (2009)

Galaxy groups and clusters

The result is that much of the baryonic matter within clusters is left in the form of hot intergalactic gas.

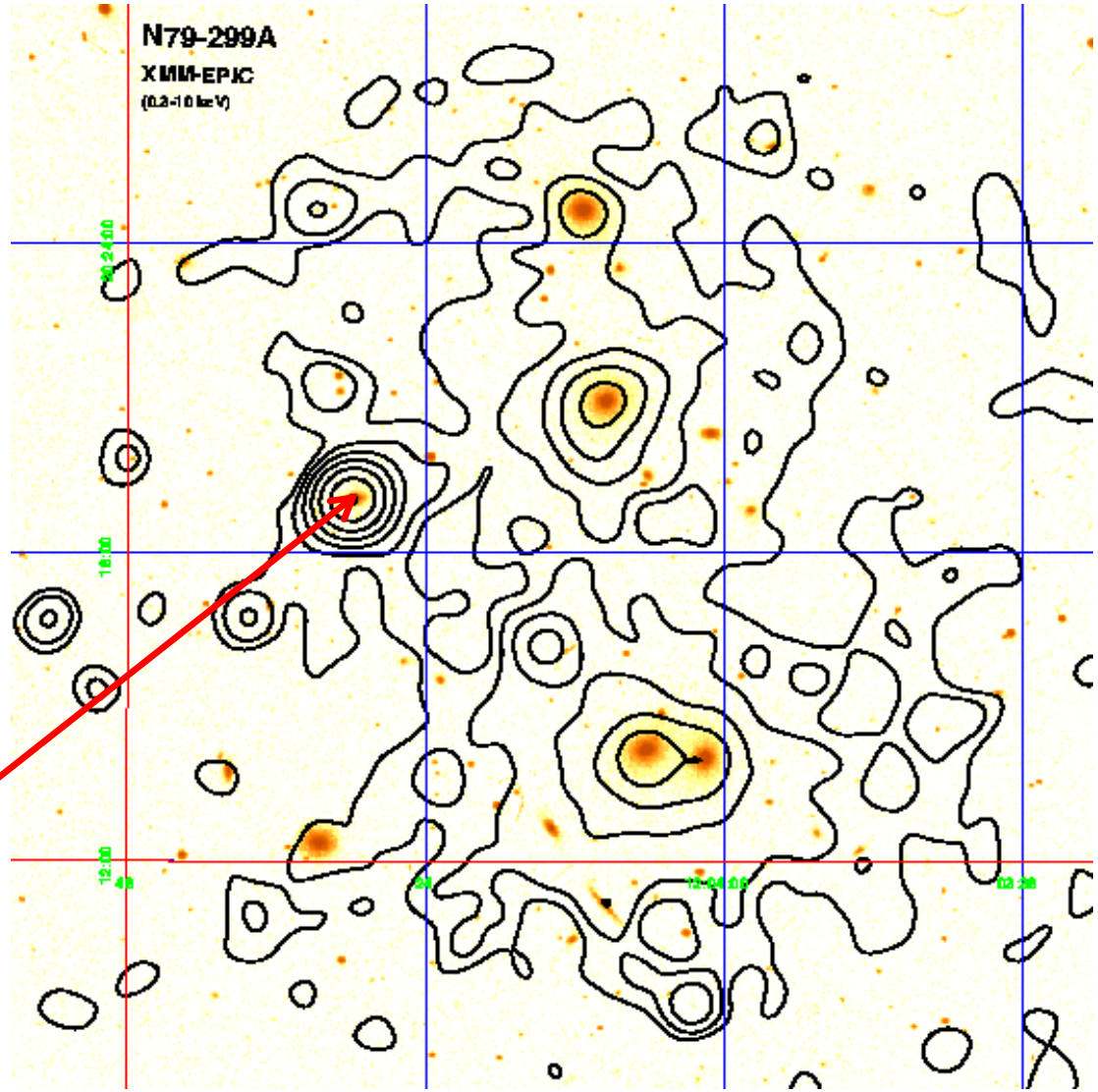




Galaxy groups and clusters

The result is that much of the baryonic matter within clusters is left in the form of hot intergalactic gas.

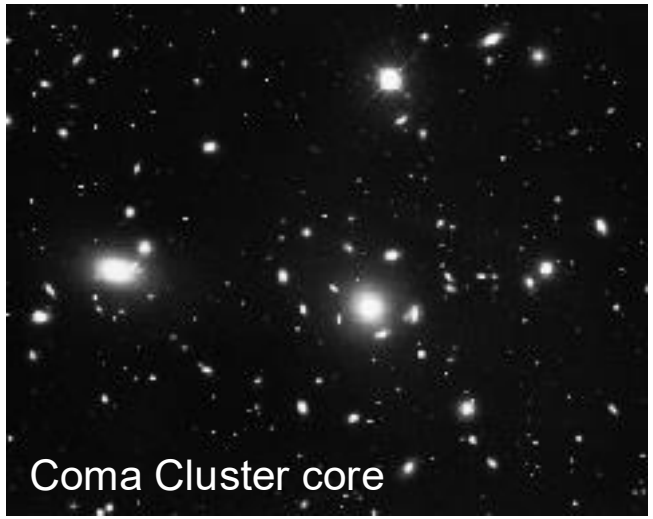
Here contours of X-ray emission from ~10^7K gas are overlaid on an optical image of the galaxies for a fairly rich galaxy group, N79-299A.



Active galaxy – strong point-like X-ray source

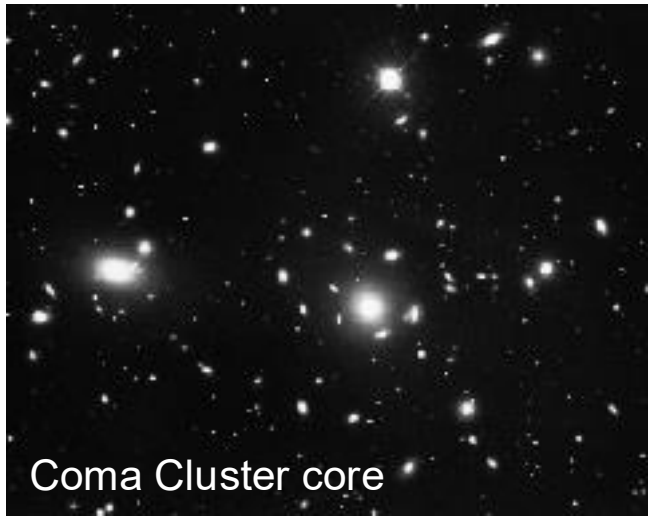
Galaxy groups and clusters

- Galaxy clusters are the largest collapsed structures in the Universe. They typically contain hundreds of luminous ($\geq 10^{10}\ L_\odot$) galaxies, and have masses $\sim 10^{14}$-$10^{15}\ M_\odot$ and radii of 1-3 Mpc.
- Only ~5-10% of galaxies are found in such rich clusters, but a much larger fraction (30-50%), including our own galaxy, lie within poorer galaxy groups with M~few $\times\ 10^{12}$-$10^{14}\ M_\odot$, and radii of a few hundred kpc.
- These systems are gravitationally bound, but not necessarily virialised.
- Abell's catalogues (1958; 1989) included 4073 rich clusters.
- X-ray studies have shown that most of the baryonic material within clusters, and at least some groups, is in hot gas.

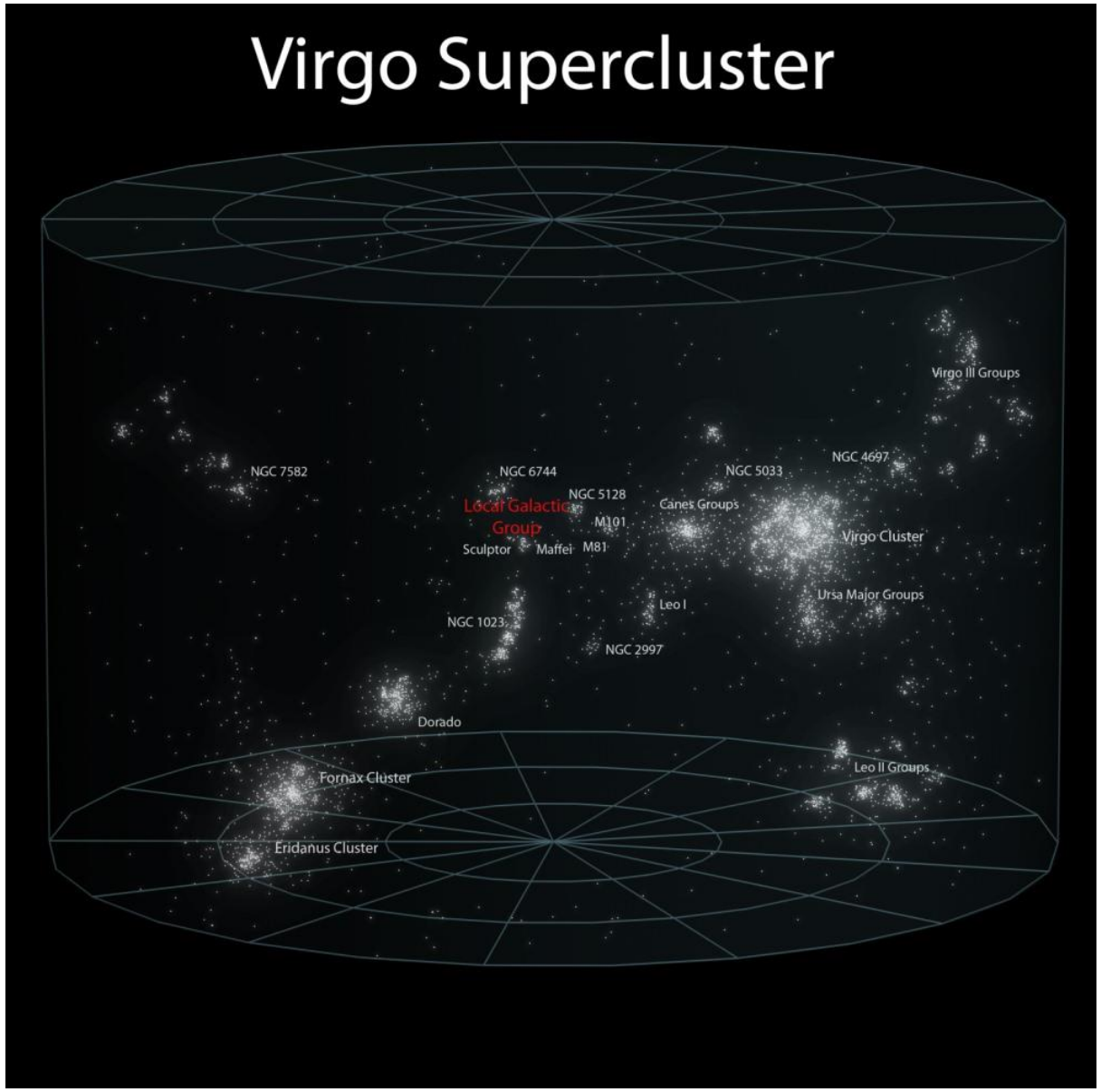


Galaxy groups and clusters

- The distinction between groups and clusters is rather arbitrary, but the cut is normally taken to be at $M \sim 10^{14}\ M_{\odot}$ or an X-ray temperature ~ 2keV.
- The nearest clusters are Virgo and Fornax (containing 100s of galaxies, at d=15-20 Mpc)
- The nearest really rich cluster is Coma, at d=100 Mpc. It is ~3 Mpc in radius.



Example - the Virgo cluster



Virgo dominates the local supercluster. It is ~16 Mpc from us and contains >2000 known galaxies, most of which are dwarfs.

Finding groups and clusters

Several techniques are used to find clusters and groups:

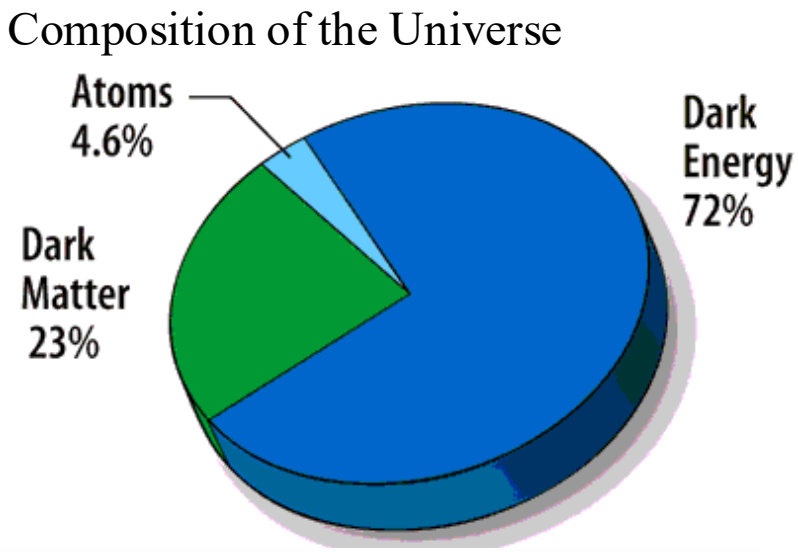
•**Optical** – search for overdensity of galaxies, either in a redshift survey (e.g. using Friends-of-Friends approach) or in projected density. This may be combined with colour selection (e.g. red sequence search). Generally use redder band (NIR) for high z clusters. Pro: plentiful deep data. Con: vulnerable to projection effects.

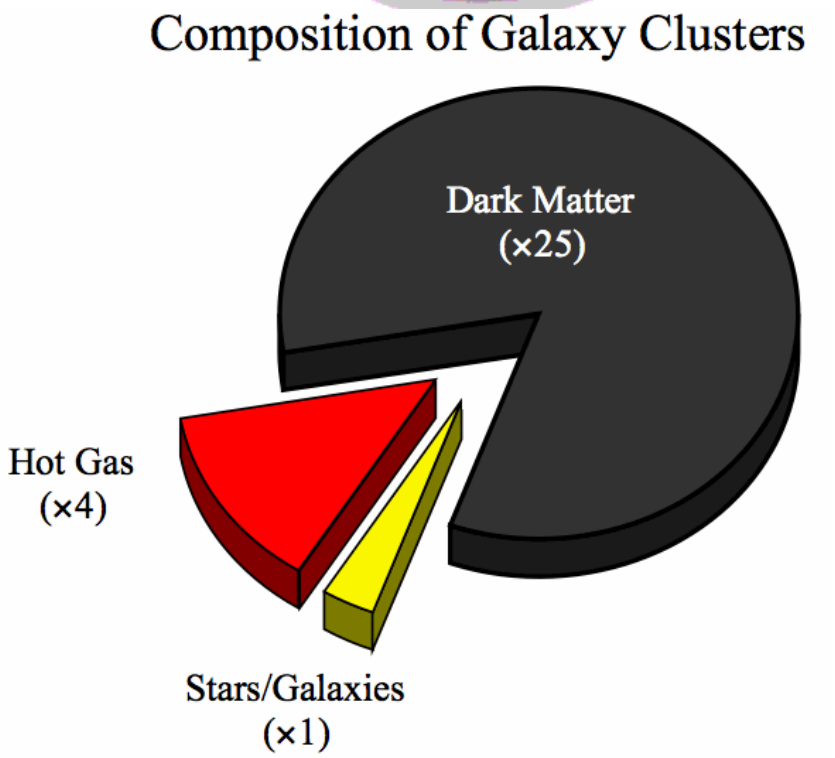
•**X-ray** – search for extended emission from hot gas. Pro: emissivity scales as ρ_{gas}^2, so projection is less problematic. Con: confusion from AGN emission, not a direct measure of cluster mass.

•**Sunyaev-Zeldovich effect** – distortion of CMB emission due to interaction with hot gas. Pro: surface brightness of the distortion does not fall off with redshift, so should be good for distant clusters. Con: only feasible for $M>10^{14}\ M_\odot$ clusters.

•**Weak lensing** – detect small distortions in background galaxies. Pro: direct detection of cluster mass. Con: challenging technique, vulnerable to projection.

Global properties



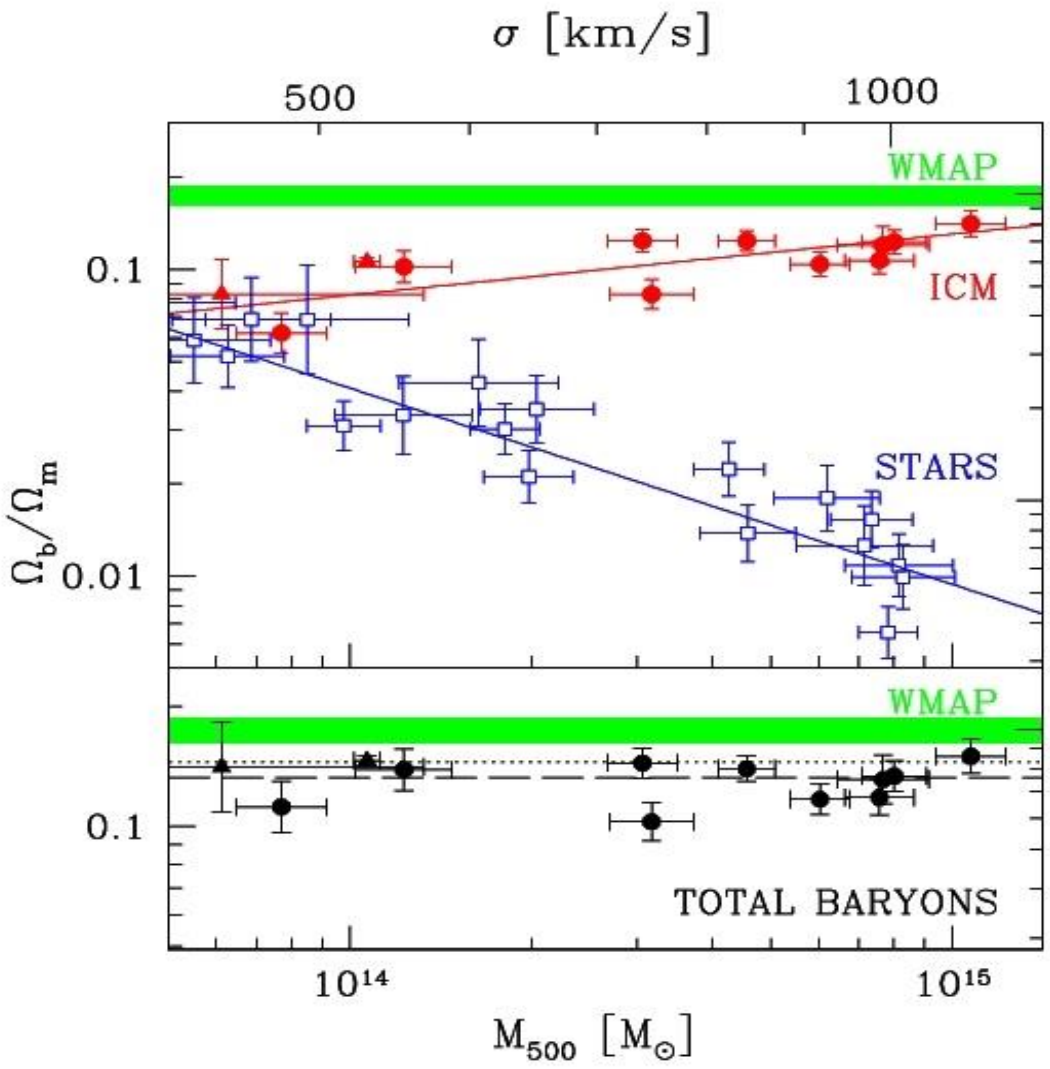


Large clusters should provide a "fair sample" of the matter in the Universe which is able to cluster – i.e. the baryonic and dark matter.

From WMAP/Planck and other constraints, this amounts to ~28% of the closure density.

Studies of clusters show that typically 85% of this is dark matter, and that ~80% of the *baryonic* matter is in hot gas.

Global properties



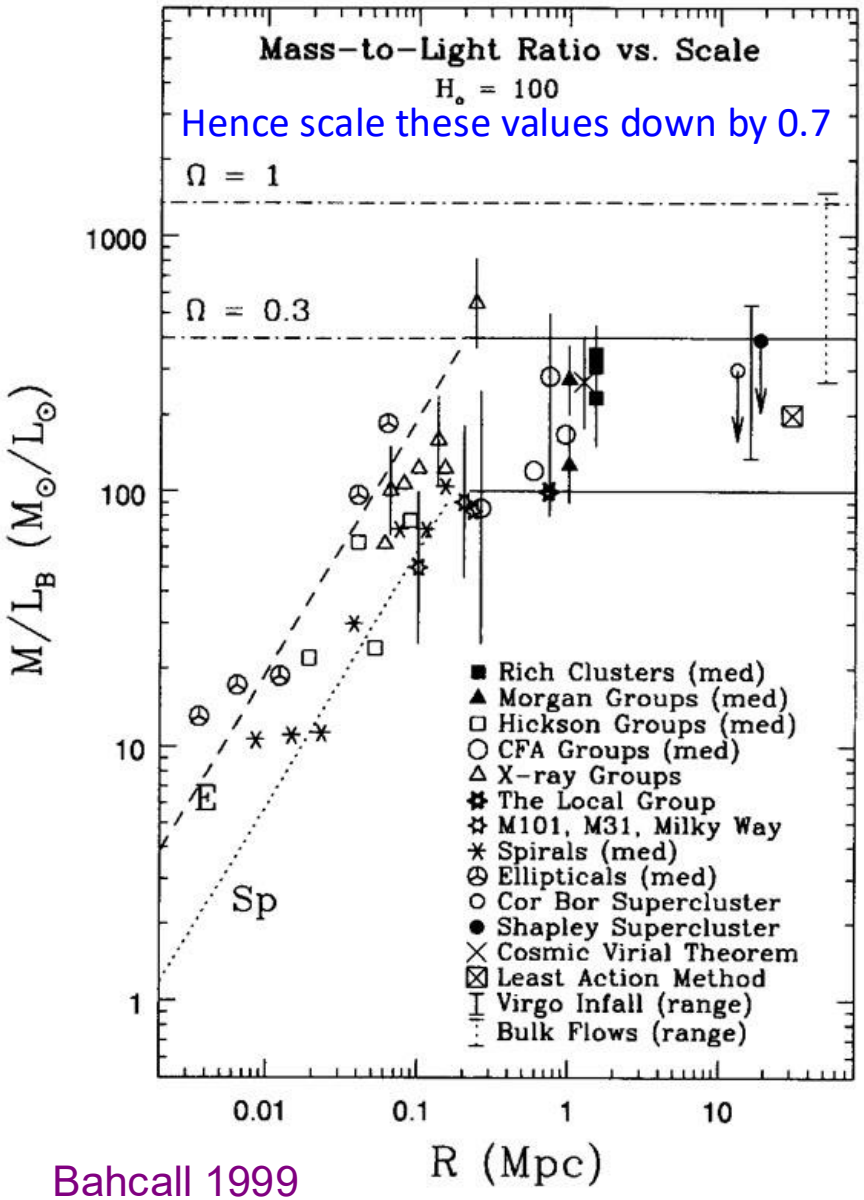
Gonzalez et al 2007

The balance between stars and gas is observed to vary substantially from groups to clusters.

The hot gas fraction rises with system mass, whilst the stellar mass fraction falls. In poor groups, the two are approximately equal.

The *total* baryon fraction seems to be rather less than the cosmic mean value of 17% (from WMAP). The issue of whether it drops in groups is still controversial.

Global properties



Bahcall 1999

The mass-to-light ratio (in the blue band) is observed to be ~200 $M_\odot/L_\odot$ in rich clusters. This can be compared with the mean value inferred for the Universe as a whole.

The mean blue band luminosity density of the Universe derived by integrating over the 2dF galaxy LF is

$$j_B = 1.7 \times 10^8\, h_{70}\, L_\odot\ \mathrm{Mpc}^{-3}$$

and the critical density of the Universe is equivalent to $\rho_c = 1.35 \times 10^{11}\, h_{70}^2\, M_\odot\ \mathrm{Mpc}^{-3}$.

So, for $\Omega_m = 0.28$, we have a cosmic M/L of $0.28 \times 1.35 \times 10^{11}\, h_{70} / 1.7 \times 10^8 = 222\, h_{70}$, in solar units.

Galaxies in clusters & groups

As we discussed in Section 7, in the context of elliptical galaxies, the Virial Theorem requires that the 1D velocity dispersion, σ, of a stable gravitationally bound system of particles is related to the total mass and radius of the system by

$$\eta \frac{GMM_*}{R} = 2\frac{M_* 3\sigma^2}{2} \quad .$$

Here M_* is the mass of the "particles" – in this case the galaxies. Hence the total mass of the system (including any dark matter) can be estimated from the size and line-of-sight velocity dispersion of a cluster by $M = \frac{3\sigma^2 R}{\eta G}$.

In practice a cluster has no well-defined outer radius, and for typical clusters a useful formula for estimating the virial mass (which is approximately M_{200}) is given (see Eke et al 2004) by

$$M = \frac{5\sigma^2 R_{\rm rms}}{G} \quad .$$

Here $R_{\rm rms}$ is the rms projected separation from the centre, of the cluster galaxies.

It was the application of the V.T. to the Coma cluster which led Fritz Zwicky in the 1930s to first postulate the existence of "dark matter", since the mass he derived for the cluster was much larger than that visible in the galaxies.

Gas in groups and clusters

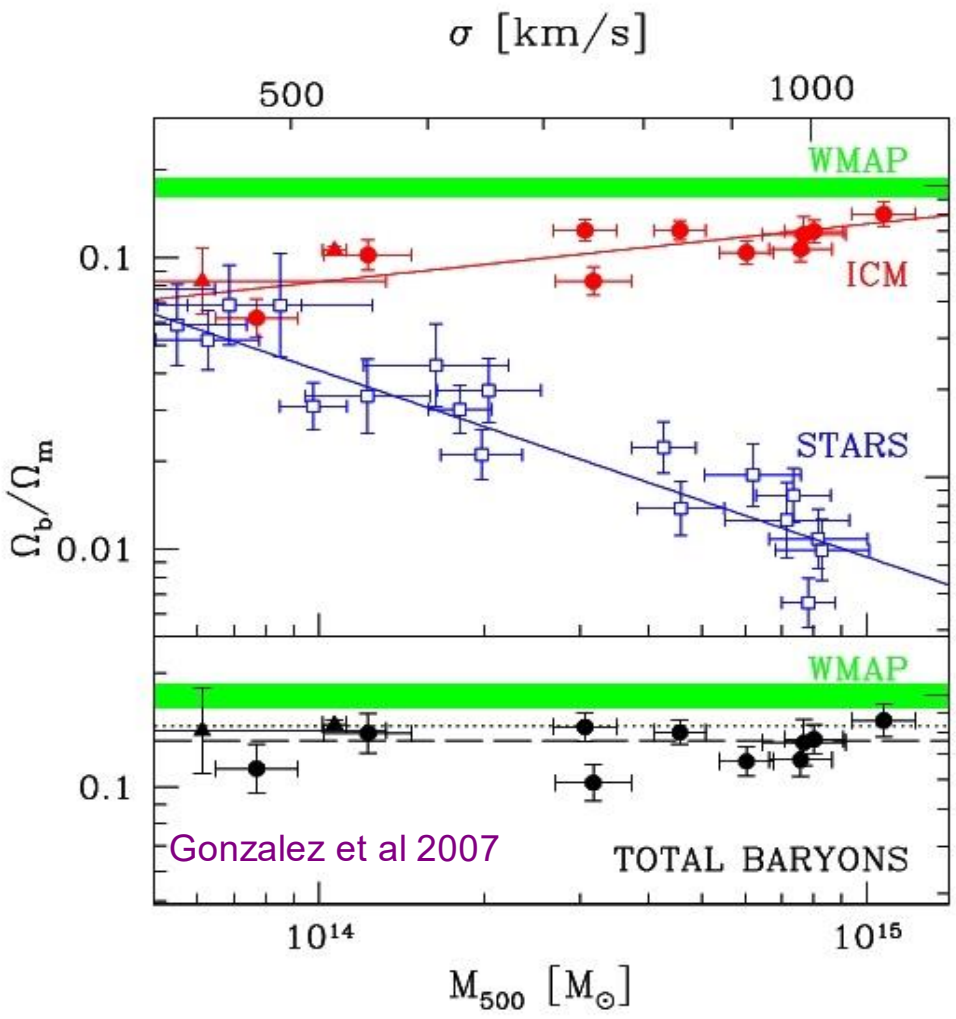
Hot intergalactic gas forms the dominant baryonic component in galaxy clusters, and in at least some galaxy groups.

The typical temperature of this gas can be estimated directly from the Virial Theorem, or by noting that it should have approximately the same specific energy as the galaxies (which are themselves subject to the VT). In other words:

$$\frac{3}{2}\frac{kT}{\mu m_p} \approx \frac{1}{2}3\sigma^2 \quad .$$

Taking μ=0.6, we find that

$$T \approx \left(\frac{\sigma}{1000\,\mathrm{km\,s^{-1}}}\right)^2 \times 6.2\ \mathrm{keV}\,.$$



Gas in groups and clusters

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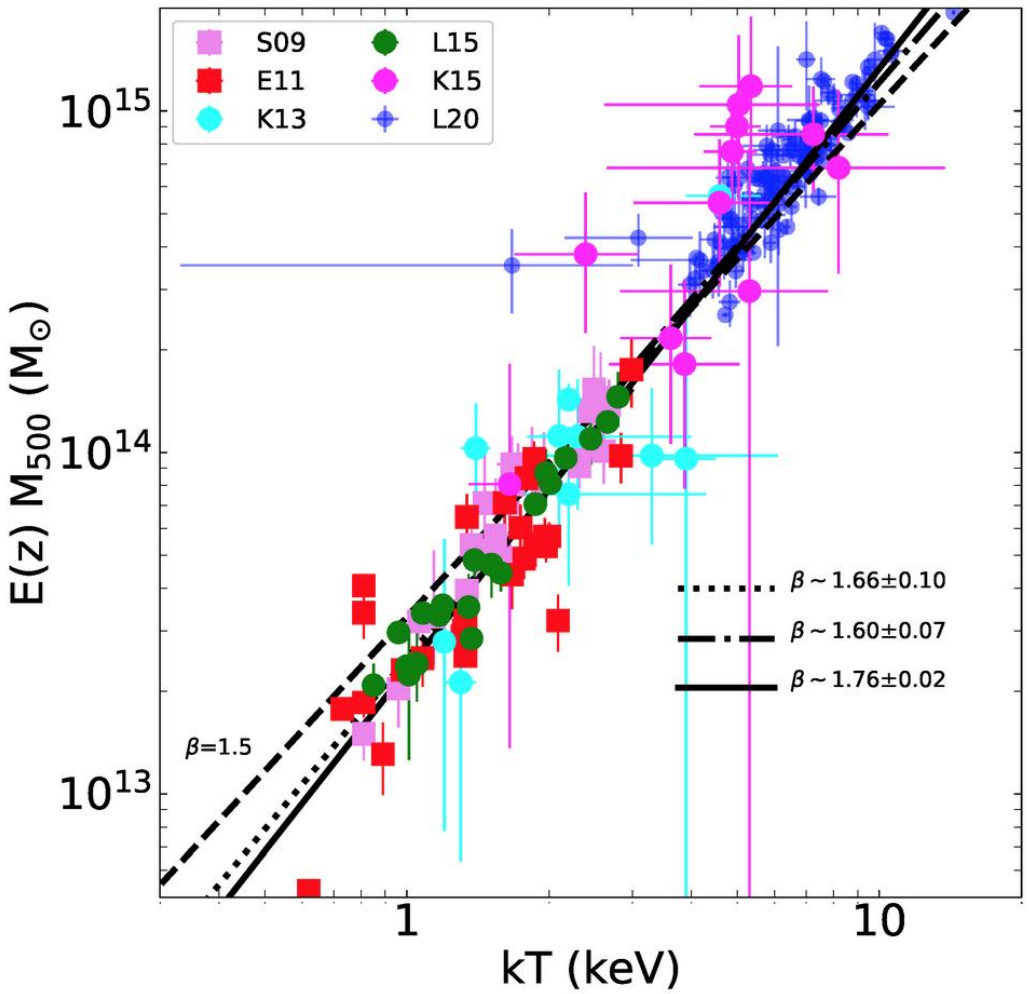

Lovisari et al 2021

N.B. 1 keV =1.16 × 10^7K.

Gas in groups and clusters

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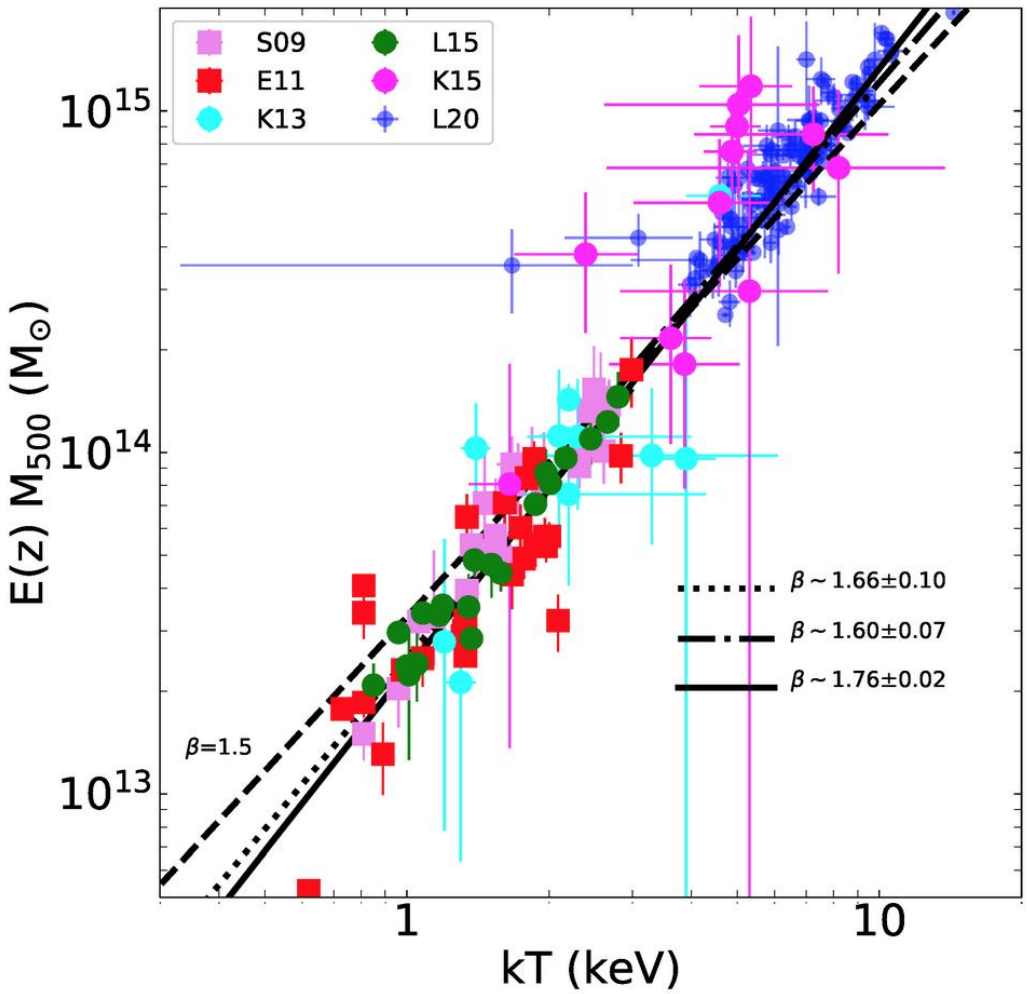
From the top hat collapse model: - all virialised systems are expected to have a mean density, within their virialised region of approximately $200\rho_c$ (where ρ_c is the critical density of the Universe at the time they virialise).

Hence for clusters which virialise at the same time, their mass and radius must be simply related by $M \propto R^3$.



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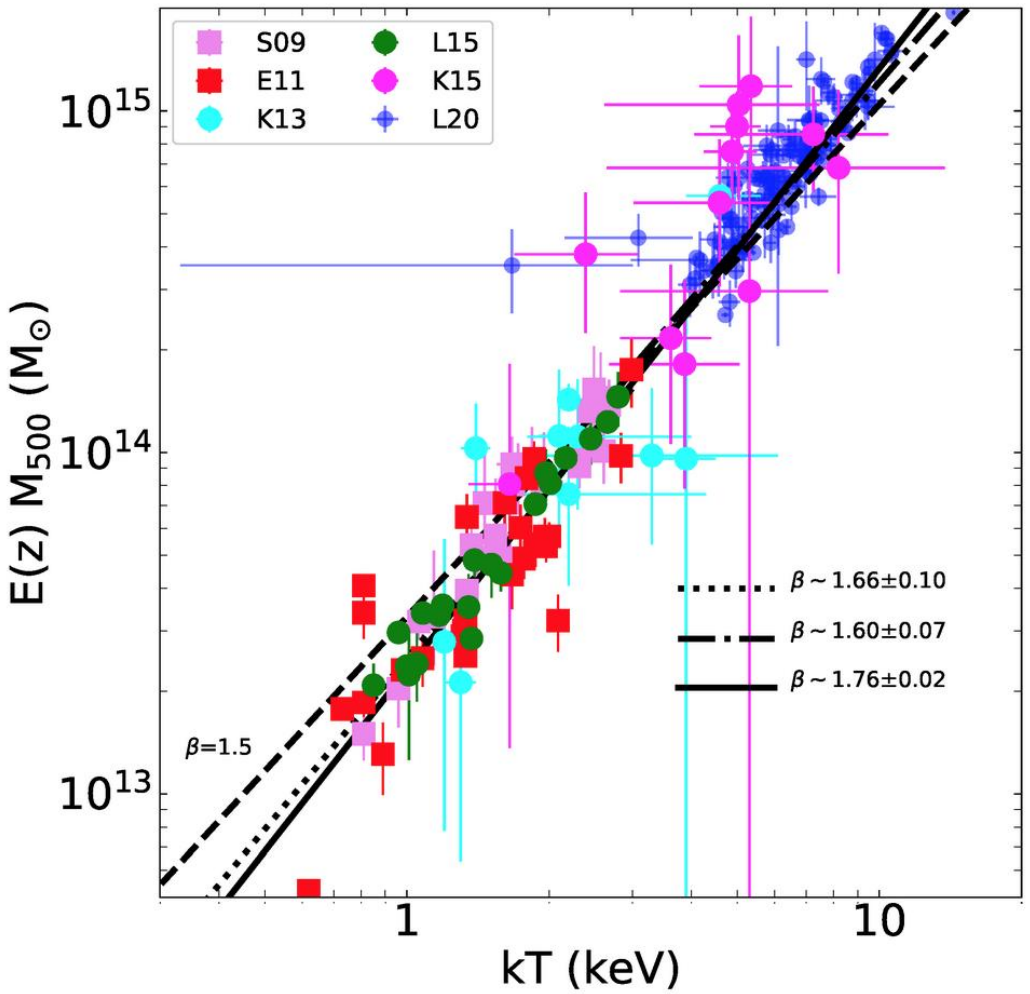
Hence for clusters which virialise at the same time, their mass and radius must be simply related by $M \propto R^3$.

It follows that $M^{2/3} \propto \sigma^2 \propto T$, and so we expect cluster mass to scale as $T^{3/2}$, which is close to what is observed.



Lovisari et al 2021

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Gas in groups and clusters

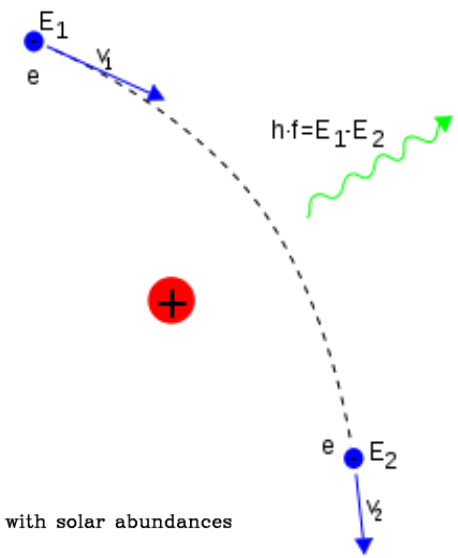
The hot gas radiates through a combination of bremsstrahlung and atomic line emission.

The bremsstrahlung emissivity has the form:

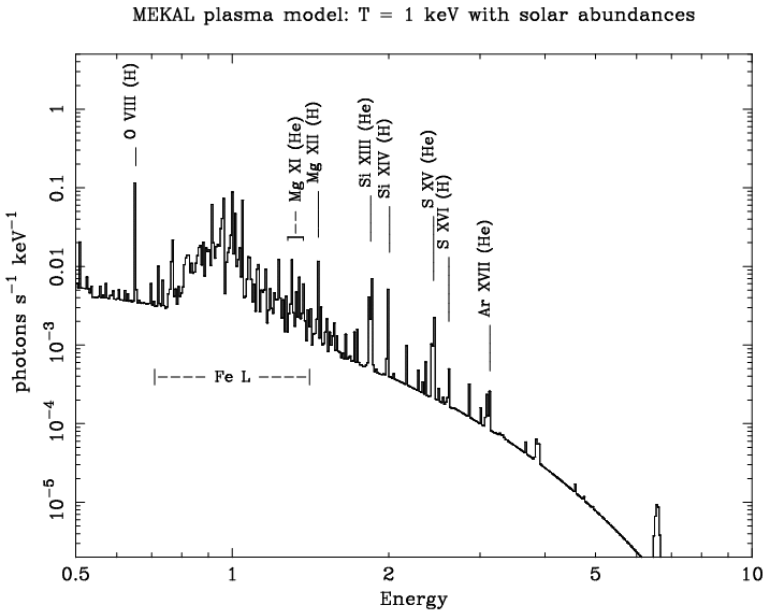
$$\epsilon(E) = A\left(\sum_i Z_i^2 n_e n_i\right) T^{1/2} g(T,E) e^{-E/kT},$$

where for gas of temperature T, the Gaunt factor is a slowly varying function of energy, which can be approximated as

$$g(T,E) \propto E^{-0.4}$$



Emission from bound-bound transitions becomes increasingly important for gas with T<3 keV, and line features within X-ray spectra are indicative of individual ionic species. Various codes (Mekal, APEC etc) are available to calculate the spectrum emerging from hot plasmas in collisional ionisation equilibrium.



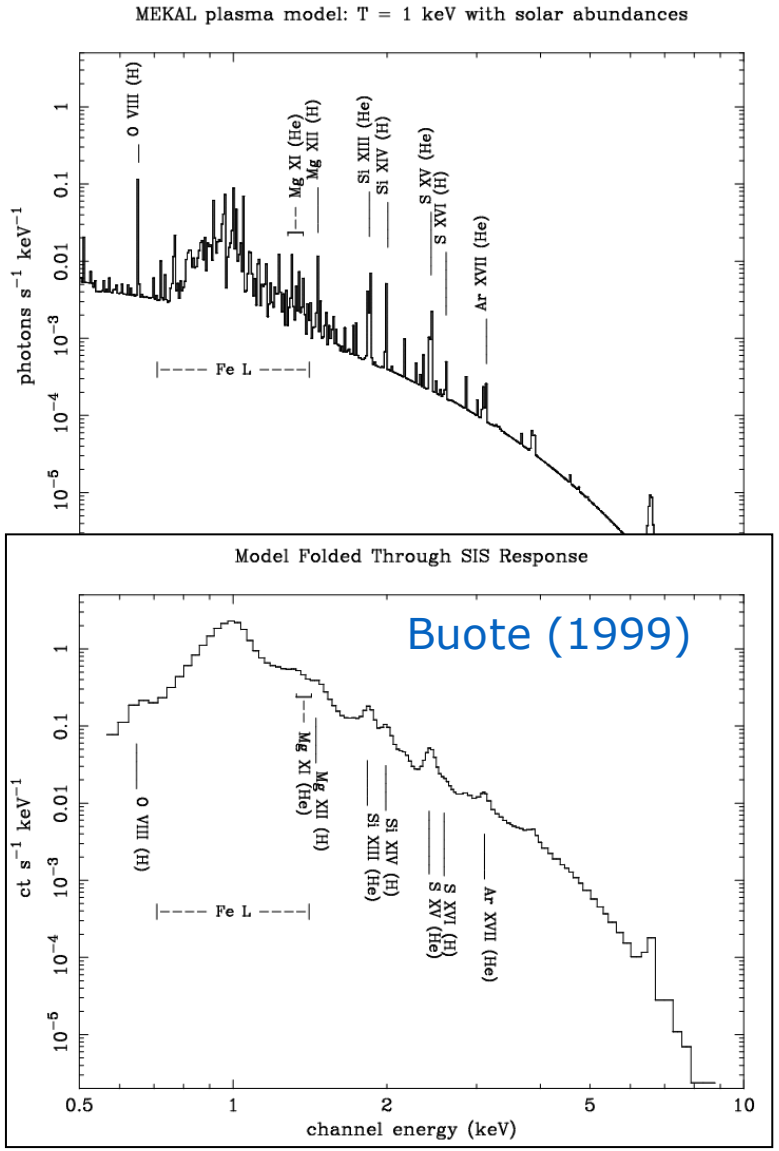
Gas in groups and clusters

In practice, the spectral resolution of X-ray spectrometers is limited, so that lines are blended together. However, it is still possible to fit hot plasma models to constrain the temperature and metallicity of the hot gas.

The total emissivity (power radiated per unit volume) of the gas can be represented as

$$\epsilon = n_e n_H \Lambda(T, Z)$$

where $\Lambda(T, Z)$ is the cooling function of gas of temperature *T* and metallicity *Z*.



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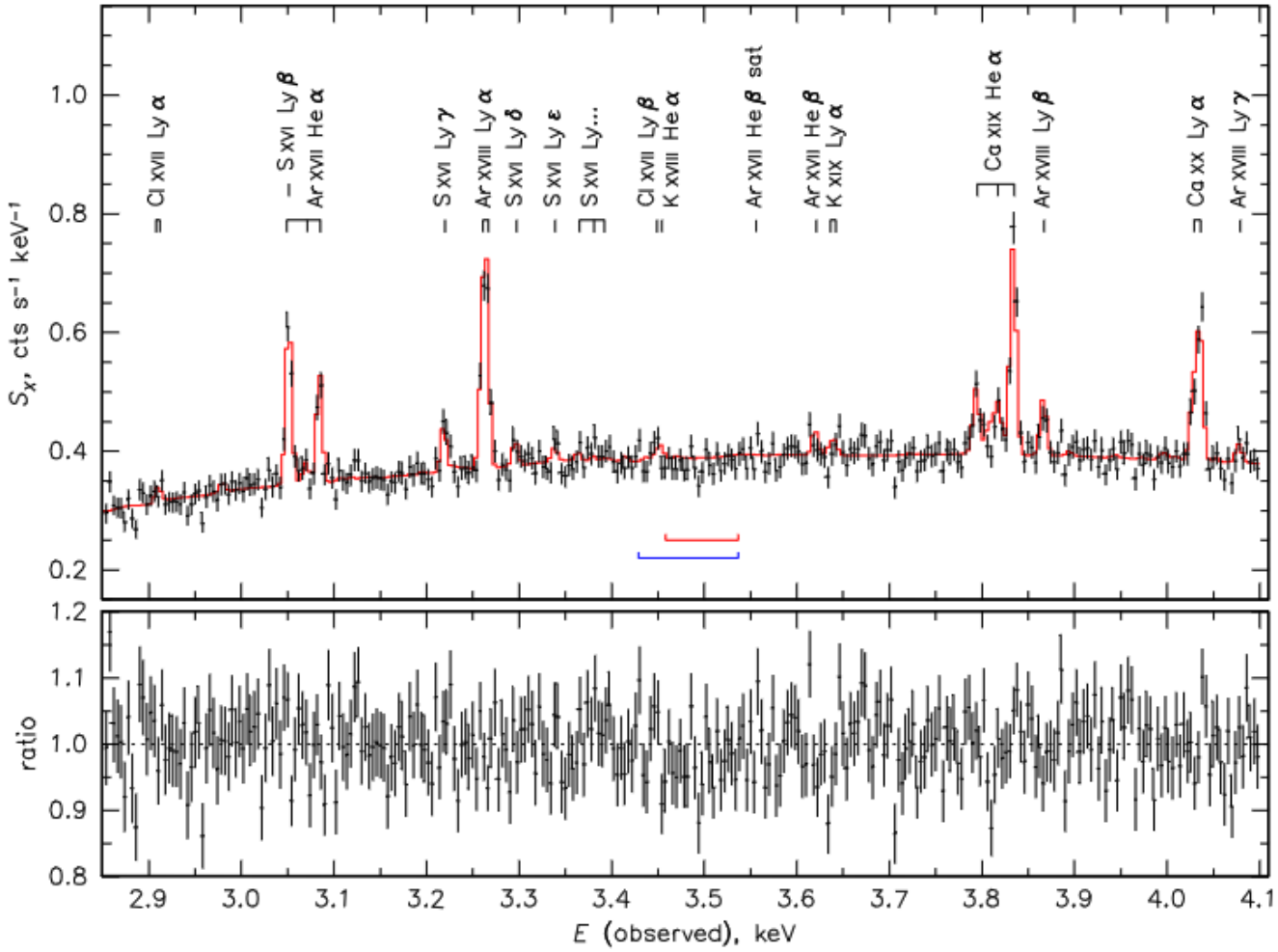
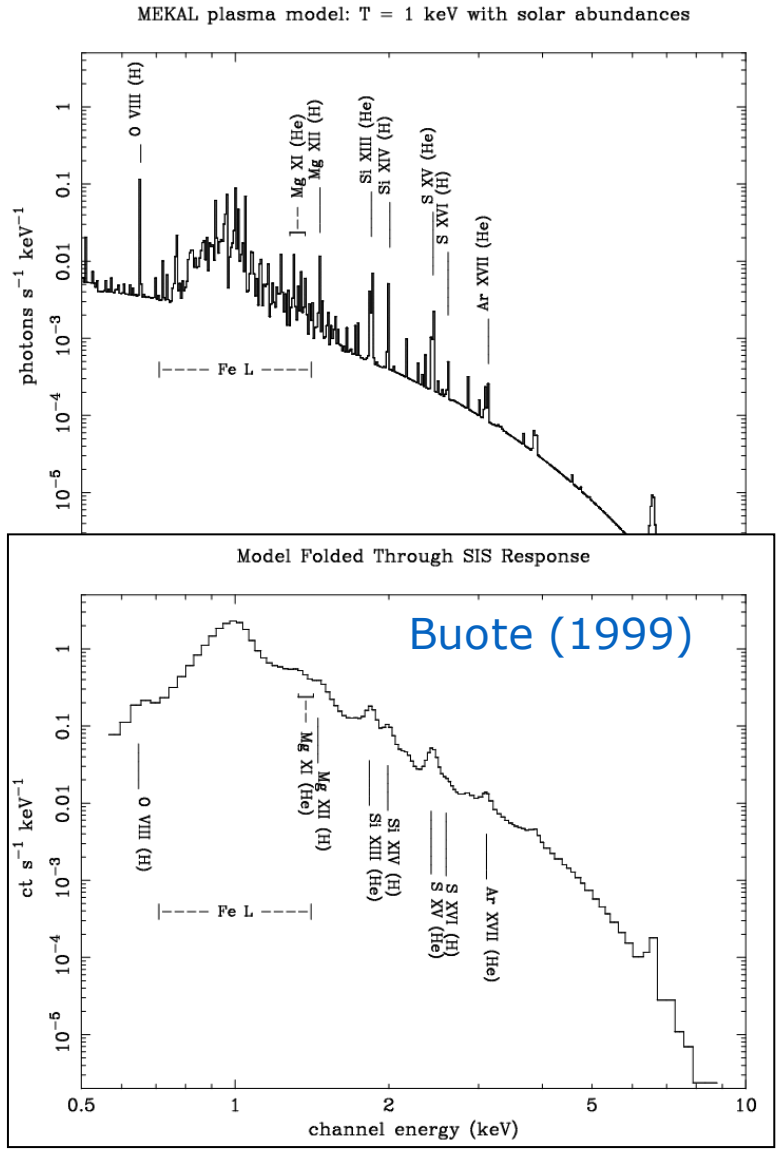


Hitomi – Aharonian et al. (2017)



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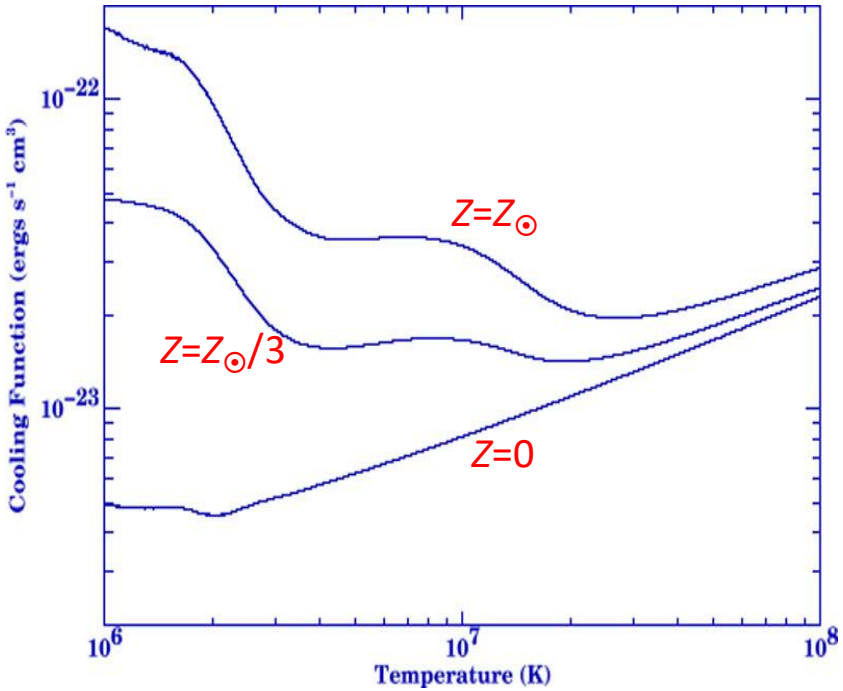
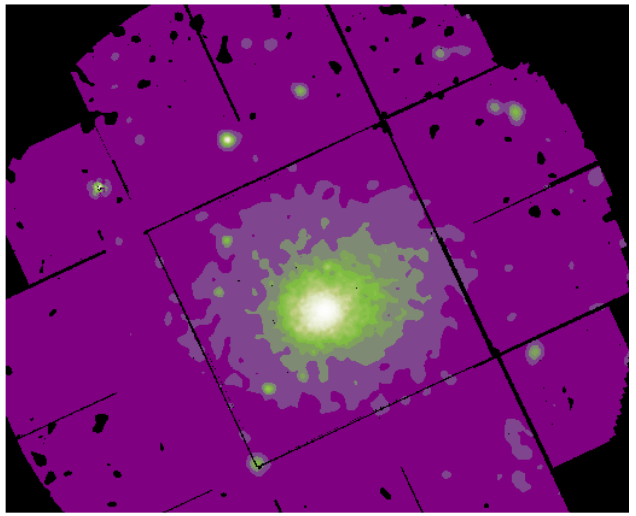
where $\Lambda(T, Z)$ is the cooling function of gas of temperature *T* and metallicity *Z*.

$\Lambda(T) \propto T^{1/2}$ for bremsstrahlung, which dominates for gas with *T*>3 keV. However, for cooler gas with significant metal content, line emission causes $\Lambda(T)$ to flatten, and even to *rise* towards lower temperatures.



X-ray mass analysis

The distribution of the X-ray emission form the hot gas in a cluster can be used to derive not only the distribution of hot gas, but also that of the total gravitating mass, and hence the dark matter. The basic approach is to use the equation of hydrostatic equilibrium $\frac{dP}{dr} = -\frac{GM(<r)\rho}{r^2}$ to infer the mass profile from the pressure gradient.



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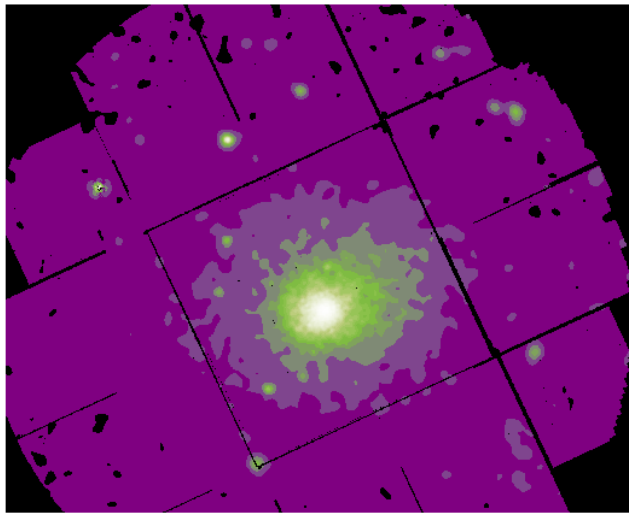
The ideal gas law is: $$P = \frac{\rho k T}{\mu m_H}$$

Inserting the ideal gas law into the equation for hydrostatic equilibrium and differentiating gives:

$$\frac{d}{dr}\left(\frac{\rho k T}{\mu m_H}\right) = \frac{k}{\mu m_H}\left[\rho\frac{dT}{dr} + T\frac{d\rho}{dr}\right] = -\frac{GM(r)}{r^2}\rho$$



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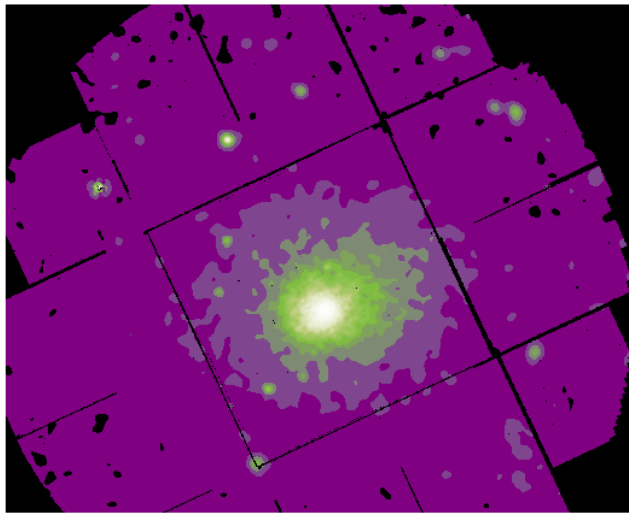

Inserting the ideal gas law into the equation for hydrostatic equilibrium and differentiating gives:

$$\frac{d}{dr}\left(\frac{\rho k T}{\mu m_H}\right) = \frac{k}{\mu m_H}\left[\rho\frac{dT}{dr} + T\frac{d\rho}{dr}\right] = -\frac{GM(r)}{r^2}\rho$$

Factoring out the T and ρ and multiplying by r gives the expression:

$$\frac{kT\rho}{\mu m_H}\left[\frac{r}{T}\frac{dT}{dr} + \frac{r}{\rho}\frac{d\rho}{dr}\right] = -\frac{GM(r)}{r}\rho$$

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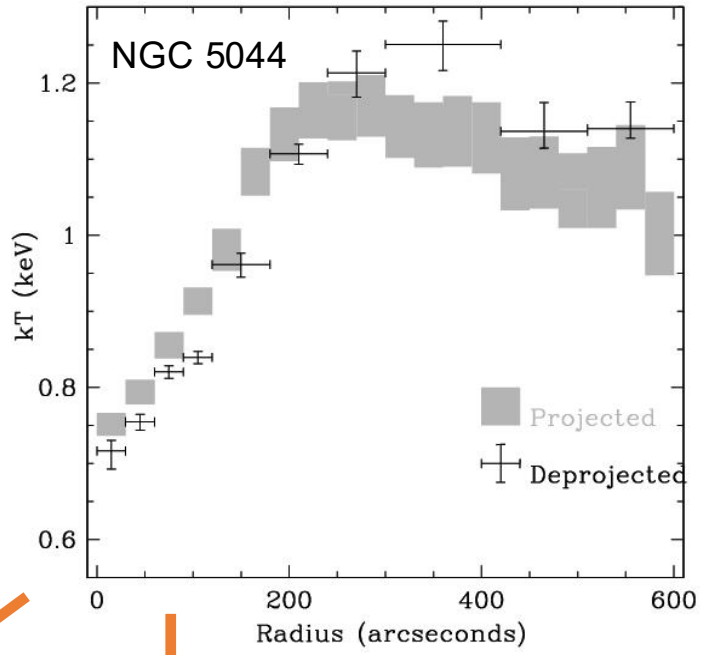
Rearranging the expression:

$$M(r) = -\frac{k_{\rm B}Tr}{\mu m_{\rm H}G}\left[\frac{{\rm dln}\rho_{\rm gas}}{{\rm dln}r} + \frac{{\rm dln}T}{{\rm dln}r}\right]$$

X-ray mass analysis

Temperature as a function of 3D radius in the system is typically derived from a deprojected fit to spectra from a set of concentric annuli.

A smooth function may then be fitted through the deprojected *T* values, to give *T(r),* and allow the log derivative to be calculated.



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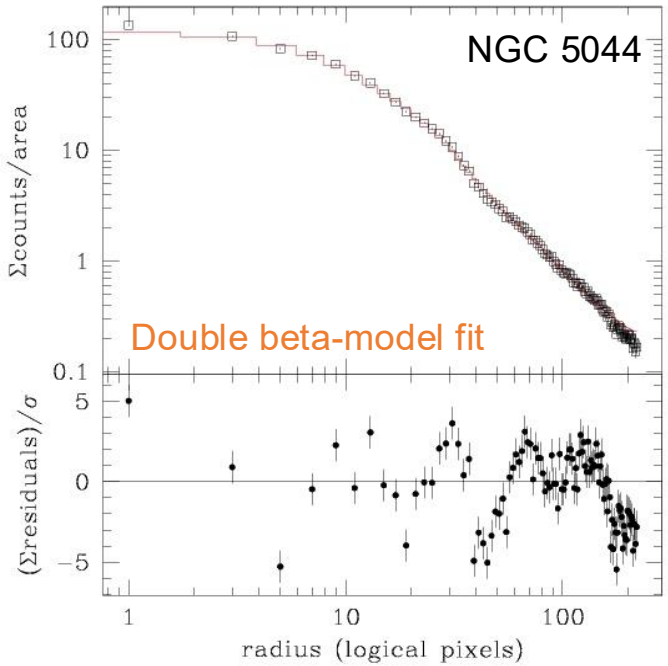
X-ray mass analysis



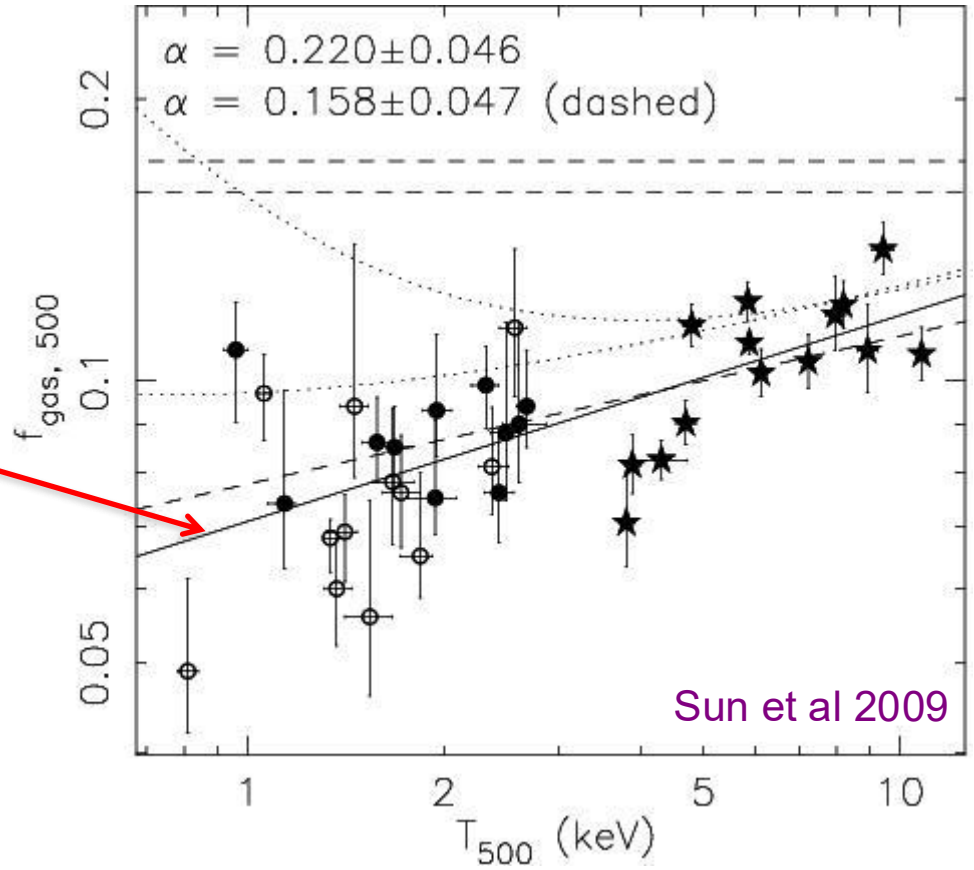
Gas density may be derived by fitting the deprojected densities inferred from the the spectral fits. For better resolution, it can be derived from a fit to the X-ray surface brightness profile, with correction for $\Lambda(T,Z)$ if necessary.

$$M(r) = -\frac{k_{\rm B}Tr}{\mu m_{\rm H}G}\left[\frac{{\rm dln}\rho_{\rm gas}}{{\rm dln}r} + \frac{{\rm dln}T}{{\rm dln}r}\right]$$

X-ray mass analysis

Combining the results for total and gas masses enables the gas fractions for groups and clusters to be derived. This rises with cluster mass. The best analysis (Sun et al. 2009) derived a trend

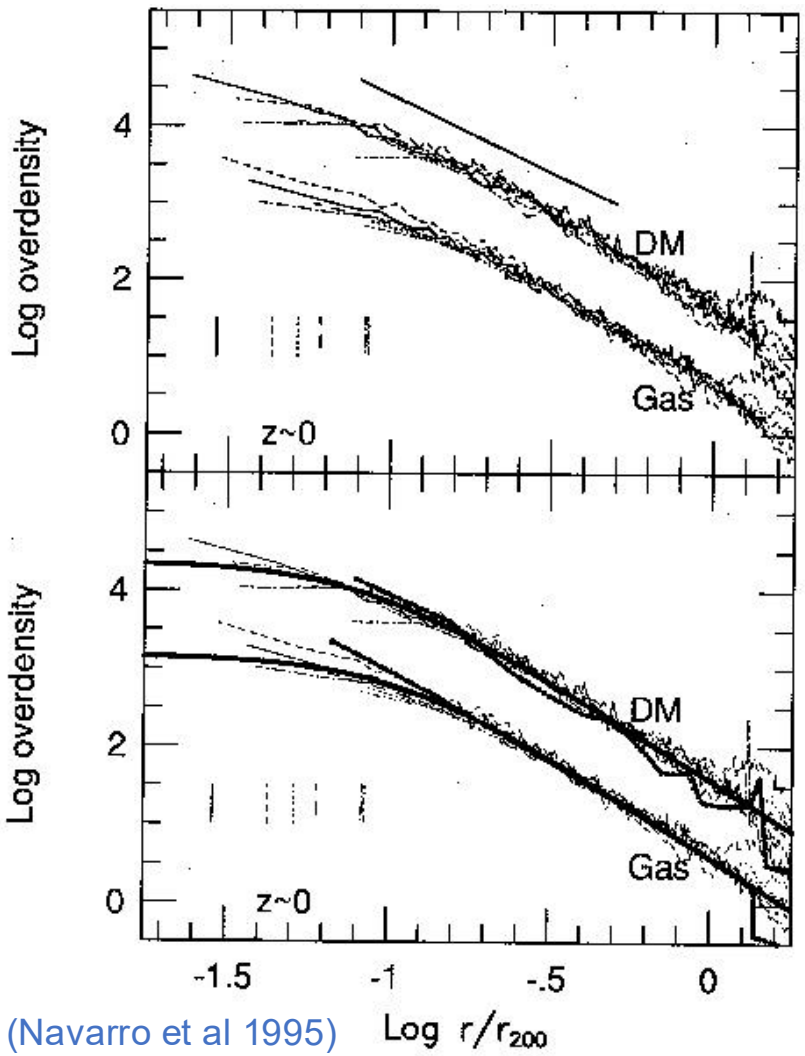
$$f_{gas} = 0.0708\ T^{0.22}$$



X-ray scaling properties

Cosmological simulations including gravity and simple gas physics produce dark halos which are almost self-similar, when scaled to a radius enclosing fixed overdensity (e.g. r_{200}).

Also, gas tracks dark matter within these halos. This behaviour would generate clusters with well-defined X-ray scaling relations.



(Navarro et al 1995)

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For fixed redshift:

$\langle\rho\rangle \propto M/R^3$ is same for all systems

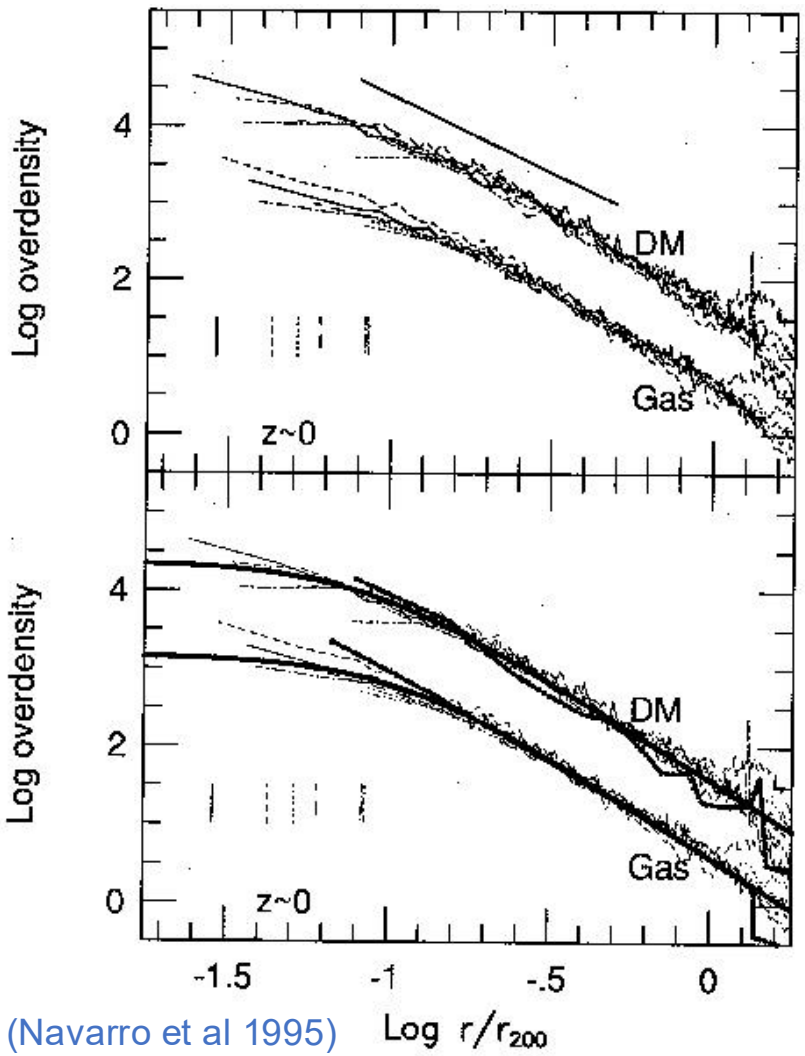
$T \propto M/R \propto R^2 \propto M^{2/3}$ from V.T.

$\therefore\ r_{200} \propto T^{1/2}$

$L_X \propto \rho^2 \cdot V \cdot \Lambda(T) \propto \rho^2 \cdot T^{3/2} \cdot \Lambda(T)$

where $\Lambda(T) \propto T^{1/2}$ for bremss,

$\rightarrow \mathbf{L_X \propto T^2}$



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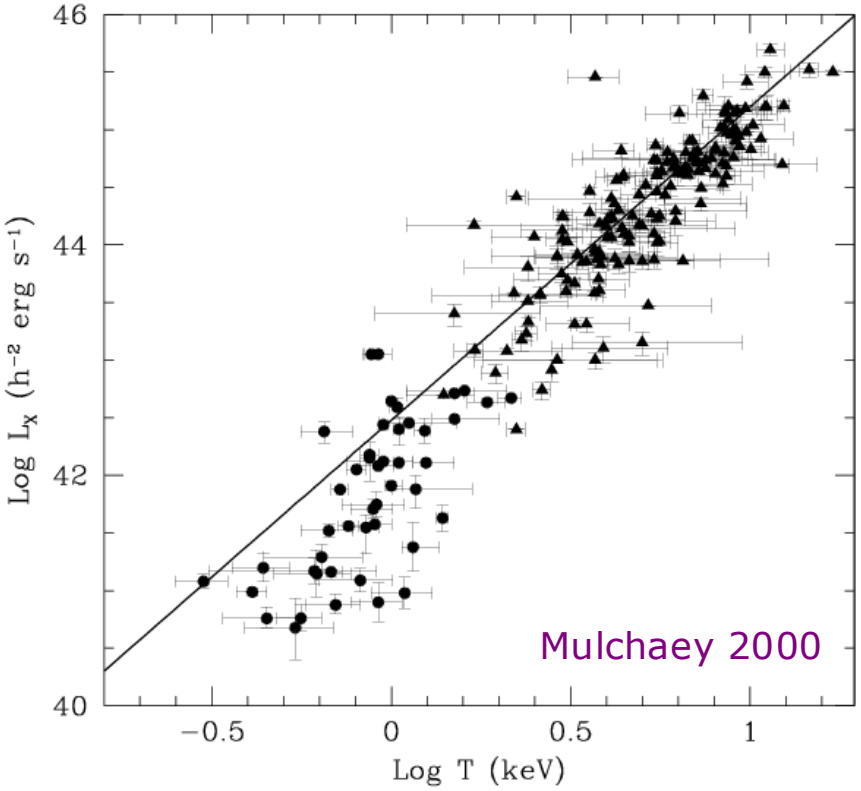
X-ray scaling properties

It has been clear for many years that the cluster L-T relation does not follow the $L \propto T^2$ slope expected for self-similar systems.

In practice, $L \propto T^{\sim 3}$ for clusters, with possible further steepening to $L \propto T^4$ in the group regime.



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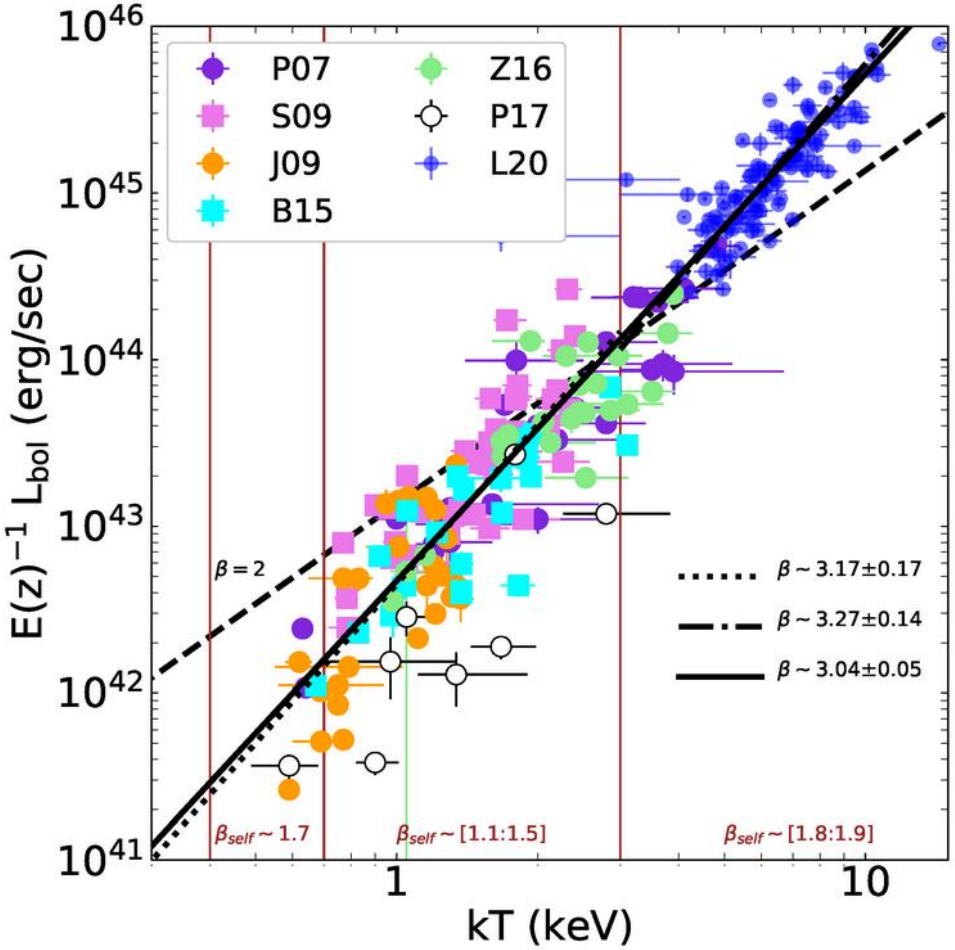
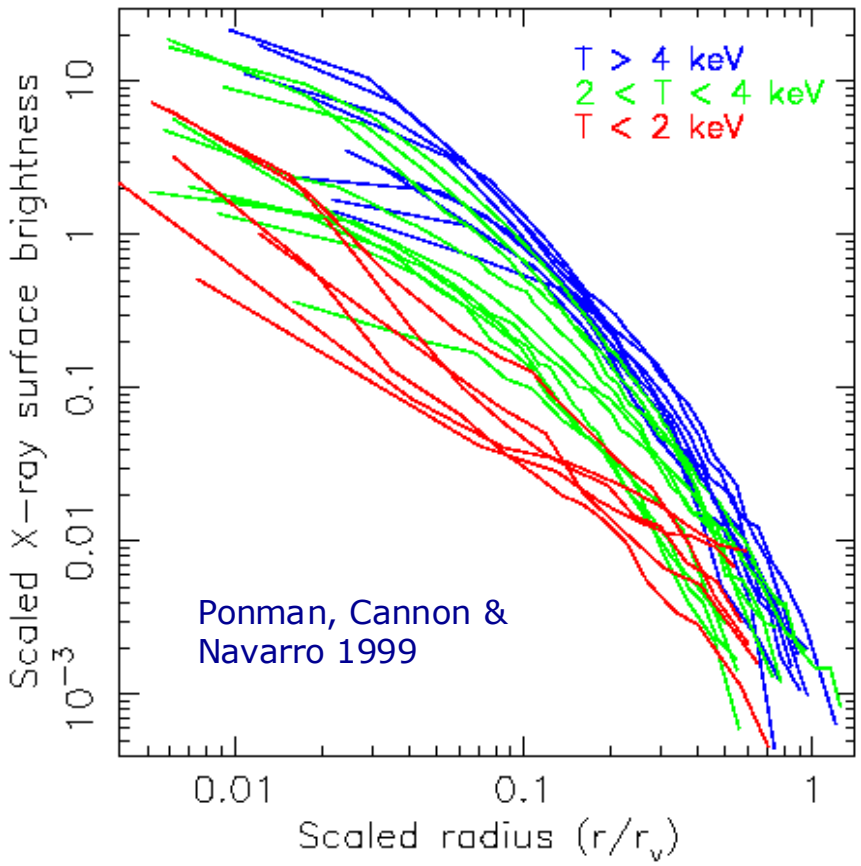


Lovisari et al 2021

X-ray scaling properties

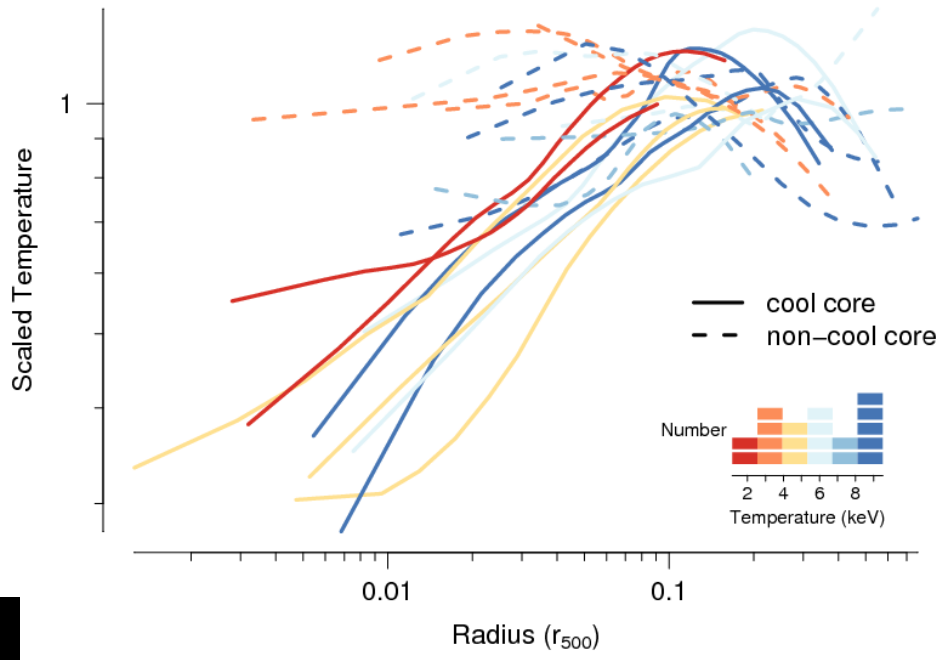
The cause of this steeper L-T relation is more obvious when one overlays scaled X-ray surface brightness profiles. This shows that emissivity (and hence ρ_{gas}) is progressively suppressed and flattened in cooler systems, relative to hot ones.

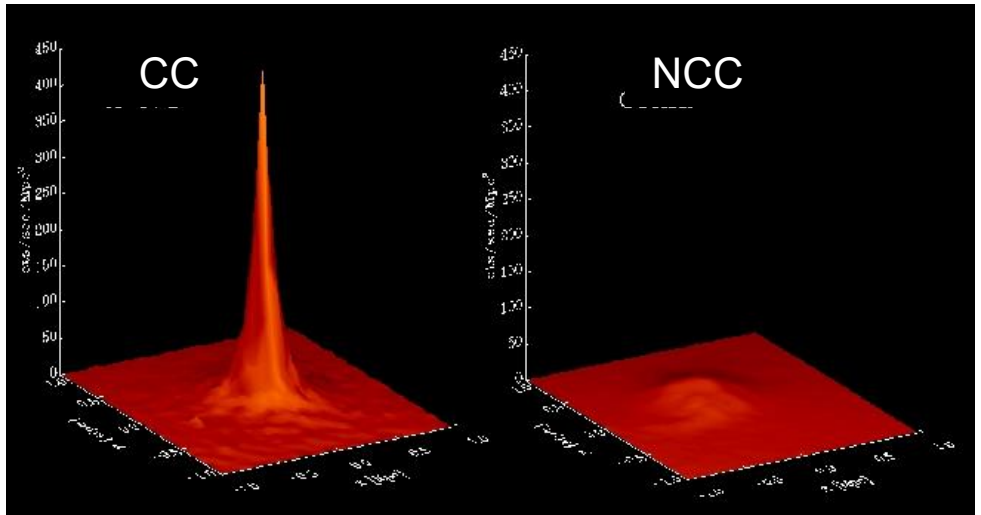
This is a sign of cosmic feedback. Injection of energy into the intergalactic gas causes it to expand in the potential well, reducing the gas density progressively in less massive cluster halos.



X-ray scaling properties

Another indication of the effects of feedback comes from the study of cluster cores. Since emissivity scales as ρ_{gas}^2, whilst thermal energy per unit volume scales as ρ_{gas}, it follows that the cooling time of the gas, $t_{cool} \sim nkT/n^2 \Lambda(T)$, is shortest in the cluster centre, where ρ_{gas} is highest. Cool core (CC) clusters show the effects of gas cooling in their cores, whilst non-cool core (NCC) ones do not.





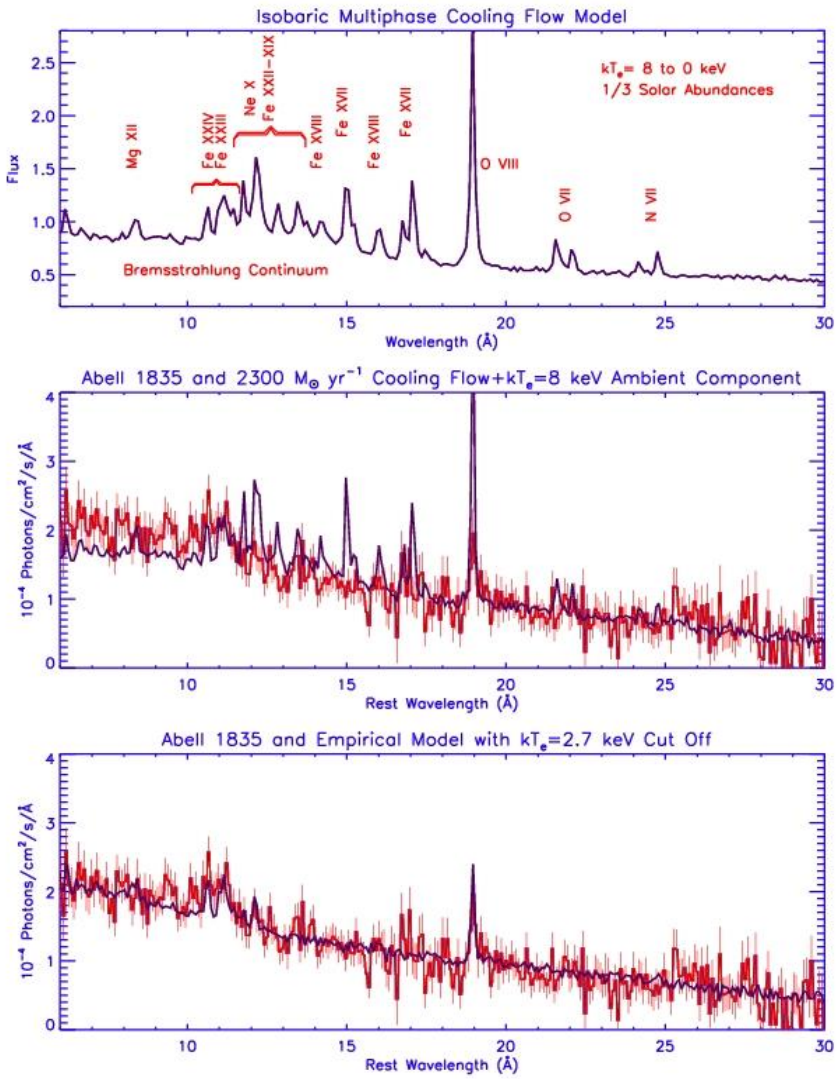
Since cooling leads to gas compression, and hence higher central emissivity, CC clusters have very strongly peaked X-ray surface brightness profiles.

X-ray scaling properties

The high resolution X-ray spectroscopy available from the Reflection Grating Spectrometer on the XMM observatory provides detailed spectral diagnostics of compact sources, including cuspy cluster cores.

This led to the important discovery that gas in cluster cores does not appear to cool in any quantity by more than a factor ~3, despite often having cooling time $<<t_H$.

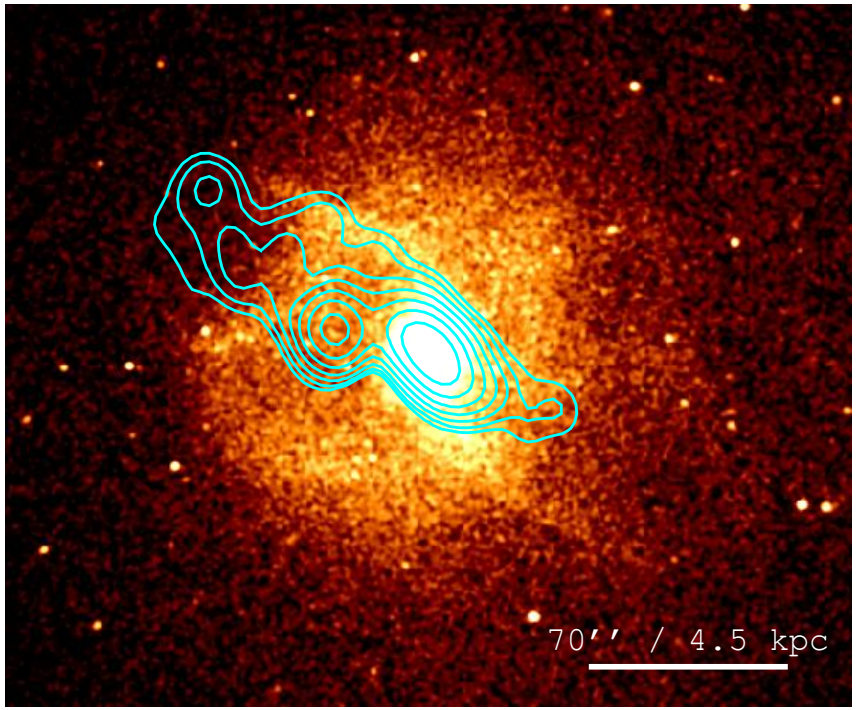
It is now widely accepted that feedback from a central AGN is responsible for greatly reducing the cooling rate of the gas. Whether this is also responsible for reducing gas density and steepening the L-T relation is still the subject of debate.



Peterson & Fabian 2006

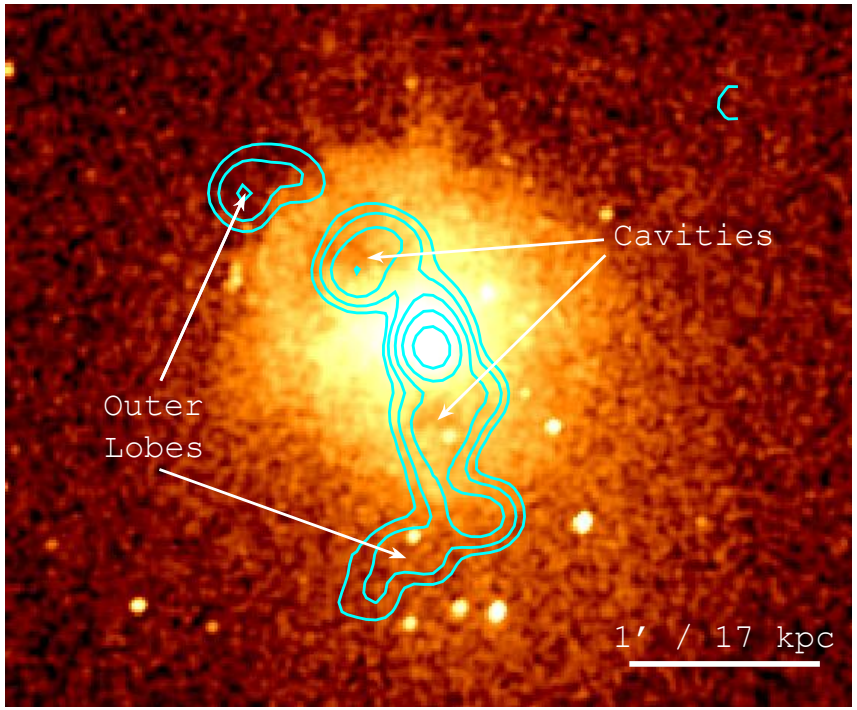
AGN feedback in groups & clusters

NGC 4636 (Baldi et al. 2009)



235 MHz contours on a Chandra X-ray image.

HCG 62 (Gitti et al. 2010)



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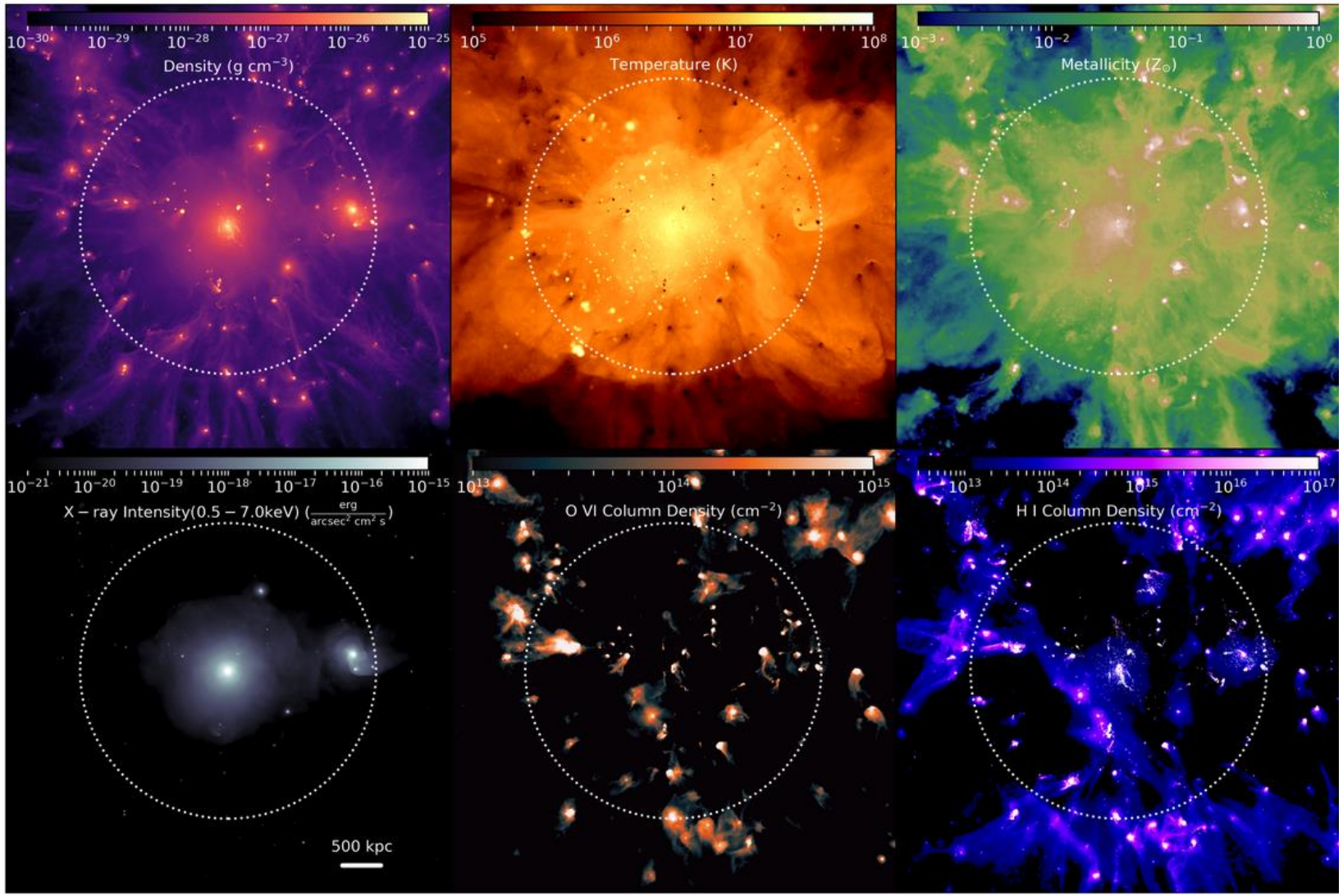
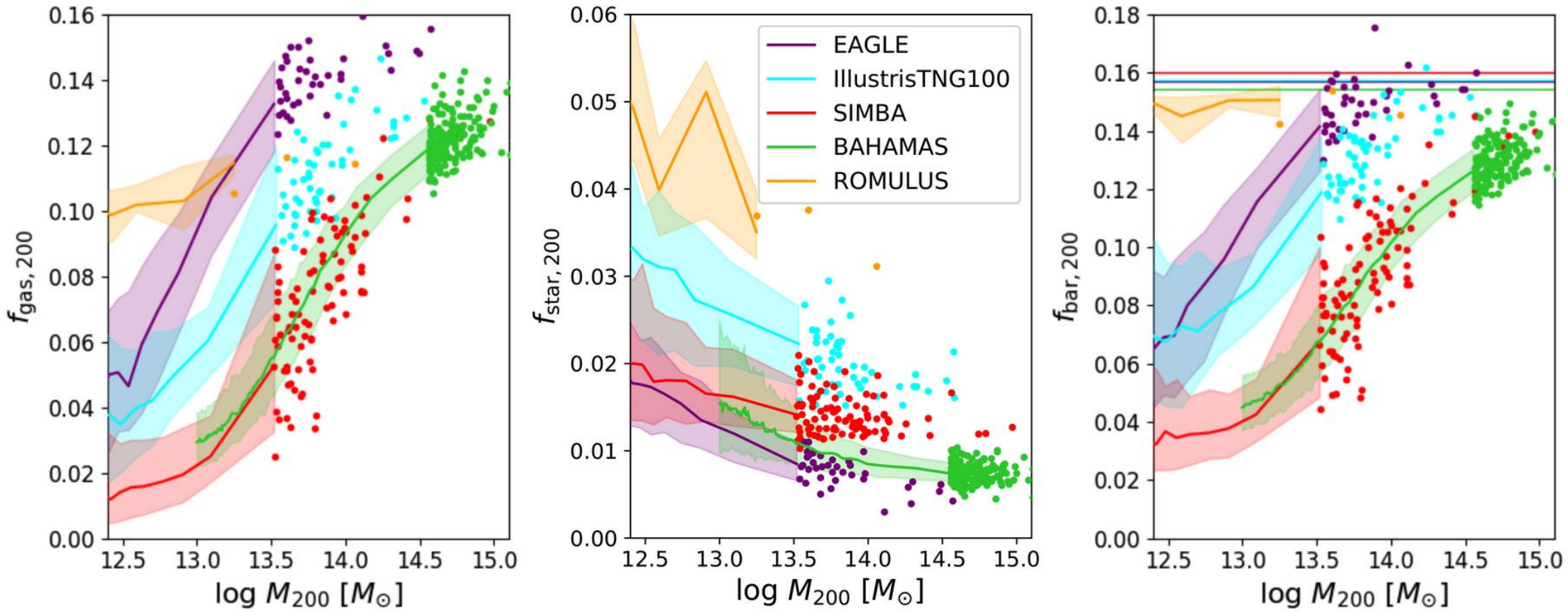
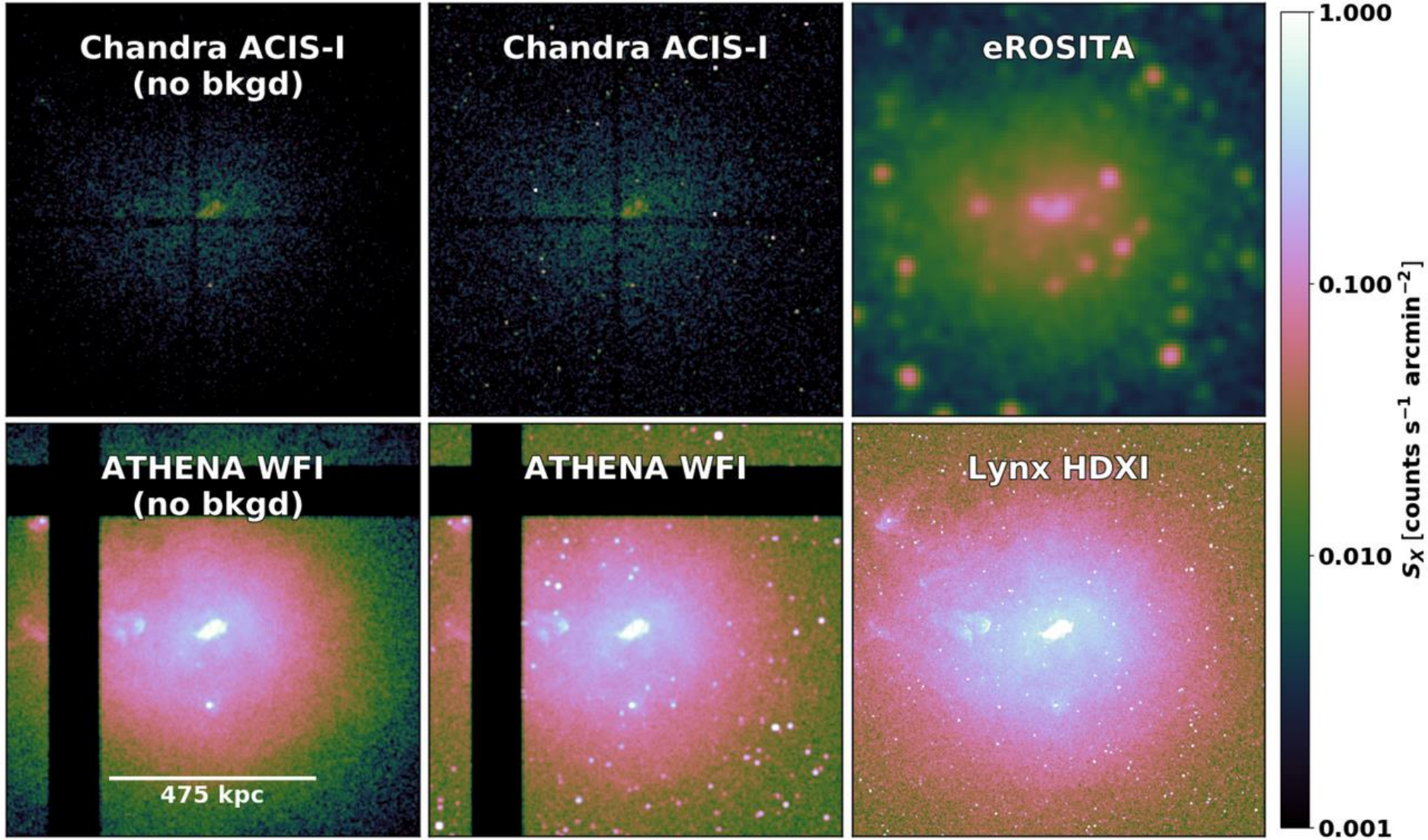
Hydrodynamical simulations of clusters



Oppenheimer et al. 2021

Drastic departures in galaxy groups



Oppenheimer et al. 2021

The future of X-ray missions